Social Security Design and Its Political Support*

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Abstract

The design of social security systems has implications for the political support of intergenerational transfers. We examine the effects of contributive vs. universal pensions, as well as the link between benefits and earnings, on the social security tax rate, labor supply, and capital accumulation. Employing an overlapping generations model featuring heterogeneous households, our analysis aligns with existing results on the effects of retirees' relative political power, ageing, and income inequality. Additionally, we find taxes to be lower when benefits are earnings-related and, if informal retirees are poorer than formal ones and the pension a comparable share of their consumption, higher when pensions are universal. Calibrated to Argentina, the model accounts for about three quarters of the rise in pension spending after the 2005–2010 expansion of coverage. Calibrated to the U.S., France and the UK, ageing is the main determinant of pension spending, the earnings link the second, and income inequality and voting patterns minor. When the electorate also chooses the design of the system, an interior choice requires redistribution through pensions to be costly; with a deadweight cost of redistribution that rises with the size of the system, calibrated to the U.S. and UK designs, ageing makes benefits more earnings-related, inequality moves pension spending little, and each of three specifications of the cost orders the designs of the UK, the U.S. and France as observed.

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1 Introduction

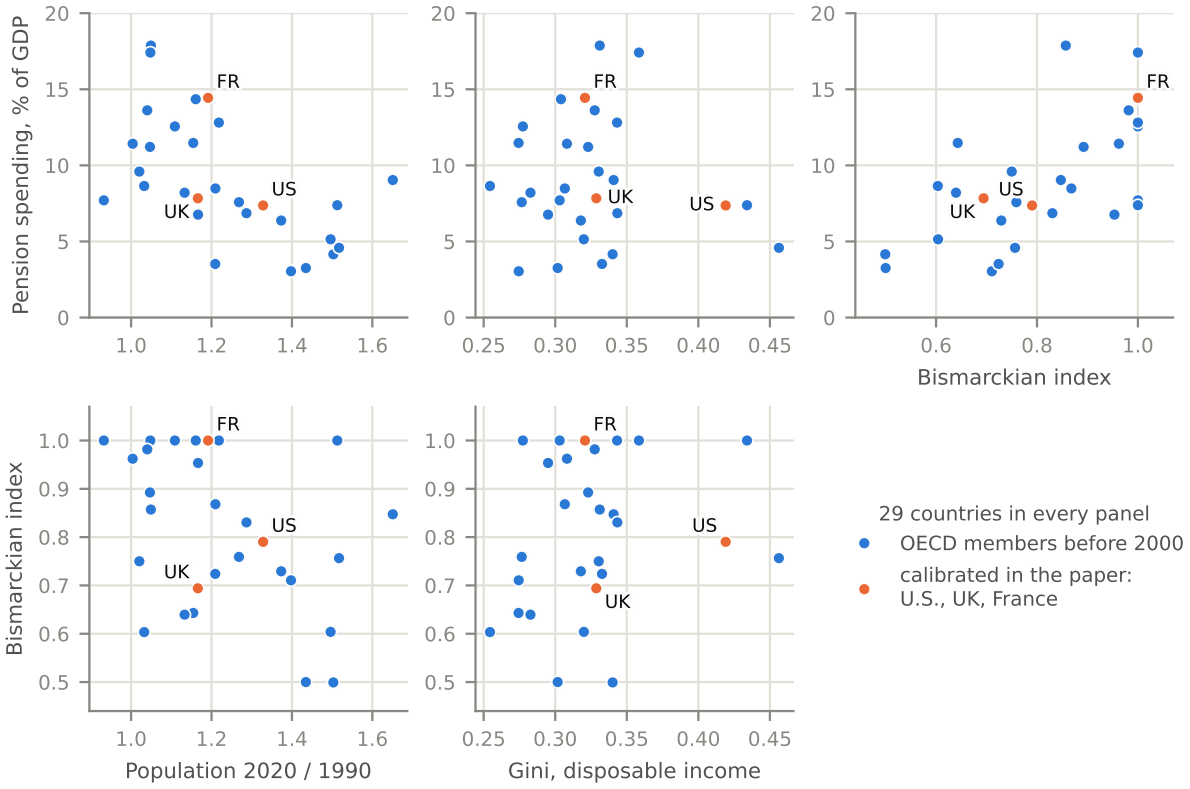
Pensions are the largest transfer programme of most governments. In 2020 public pension spending averaged about 8% of GDP across the OECD and close to 4% in emerging and middle-income economies.[1] Systems differ along two key dimensions. The first concerns eligibility and coverage: some condition benefits on a contribution record, others offer universal entitlements. The second is the link between benefits and past earnings: Bismarckian systems tie the two closely, whereas Beveridgean systems pay flat benefits. Both dimensions shape the labor supply of workers, the redistribution embedded in the system, and the coalition of voters that supports it.

Given their size, it is no surprise that the design of social security systems remains a central concern in policy debates. As populations age, policymakers must navigate difficult trade-offs among intergenerational equity, fiscal sustainability, and labor market efficiency. Moreover, pension arrangements are not determined by actuarial considerations alone: they are the outcome of political processes in which heterogeneous voters and organized interests shape the level and composition of benefits and contributions. A political-economy perspective is therefore critical for understanding why countries differ in the size of their systems and in their design. Figure 1.1 shows how they differ across the 29 countries that were in the OECD before 2000. Pension spending and the Bismarckian–Beveridgean index, the ratio of replacement rates at mean and at half mean earnings, both rise as population growth falls, and they rise together. Neither is related to income inequality. The last fact runs against the leading theory of pension design, that of Conde-Ruiz and Profeta (2007), in which high inequality leads a coalition of the rich and the poor to choose a small Beveridgean system and low inequality leads the middle class to choose a larger Bismarckian one, and few other politico-economic models have integrated the design. We address this gap with a dynamic politico-economic model and ask whether one political process can deliver both the size of the system and its design, and their relation to ageing but not to inequality.

We study the question in an overlapping generations model with capital accumulation and heterogeneous households. Households consume, save, and supply labor, and as voters, they choose among policies on labor income taxes and retirement benefits. Policy preferences vary by income and age; we model the resolution of these conflicts through probabilistic voting with elections taking place every period. Policy decisions not only impact economic outcomes, but also affect future policy decisions. Voters take these effects into account. We study Markov equilibria in which only fundamental state variables enter. Households differ in their productivity and propensity to vote. In addition, a share of households work in an informal sector or get income from home production,

[1] The dispersion is wide even among rich countries: a bit more than 3% of GDP in Ireland and Chile and more than 17% in Italy and Greece (OECD 2021).

Figure 1.1: Population growth, income inequality, pension spending and design



Sources: OECD Social Expenditure Database (pension spending), OECD historical population data (population growth), OECD, *Pensions at a Glance 2021*, table 4.1 (Bismarckian–Beveridgean index) and the UNU-WIDER World Income Inequality Database (Gini coefficient). Details are in the technical documentation (Berg and Gonzalez-Eiras 2026).

where labor is not taxed and savings do not enter the capital stock. Preferences are those of Greenwood et al. (1988), so that labor supply depends on the after-tax wage and how current effort raises future benefits. Benefits combine a flat component with an earnings-related one, the weight of the second being the Bismarckian index, and informal households receive a universal pension whose level is the second design parameter.

We show that when utility is logarithmic, politico-economic equilibria retain analytical tractability despite the presence of heterogeneity and institutional richness. With hand-to-mouth informal households, the equilibrium tax is a function of parameters and demographics alone. The analytical frame clarifies how population ageing, retirees' political influence, pension design, and income inequality shape the equilibrium tax rate. A more Bismarckian benefit formula lowers the equilibrium tax, since it shifts benefits from poor retirees, who value them more at the margin, to rich ones, so that retirees as a group support a smaller system. The effect of a more universal pension is ambiguous: it raises the tax when informal retirees are poorer than formal ones and the pension is a comparable share of their consumption. When utility is constant relative risk aversion (CRRA), we show that the state space for equilibrium policies can be reduced to a single

aggregate state and provide numerical solutions.[2]

We present two empirical applications of the model. First, we study Argentina as the case in which the design changed: from 2005 a series of tax amnesties raised the coverage of the elderly from 68% to over 90%, and pension spending by 1.8% of GDP. Implementing this reform in our calibrated model accounts for about three quarters of the observed rise in pension spending. The model also predicts modest declines in labor supply and savings, consistent with available evidence (Gonzalez-Rozada and Ruffo 2016). Second, initially taking the design as given, we calibrate the model to the U.S., France, and the UK, three advanced economies with distinct pension system characteristics, demographics, and inequality profiles. We find that the size of the pension system increases strongly with ageing and flat benefits and responds only little to income inequality, leisure preferences, and voting profiles.

We then let the electorate choose the earnings link as well as the tax. Without a cost of redistribution the choice is a corner. The flat component of benefits is a pure transfer within the retired cohort, from retirees above mean income to those below, and an electorate that weighs retirees by their marginal utility wants all of it: we prove that unless propensities to vote rise sharply with income, the design chosen with the tax is fully Beveridgean, whatever the tax rate, the demography and the income distribution.[3]

An interior design requires flat benefits to be more expensive than earnings-related ones. We introduce a deadweight cost of the redistribution the flat component performs, a leaky bucket in the sense of Okun (1975) whose leak has two parts, a constant share of every unit carried and a share that rises with the load, both scaled by the size of the system. Of three variants, the linear-quadratic cost has both parts, with two parameters calibrated so that the U.S. and UK electorates choose their observed designs; France is the test. Every variant orders the three designs as observed, but under the linear-quadratic cost France's fully Bismarckian design would need a larger constant share than the UK's design allows. With this formulation, an ageing population chooses both a more Bismarckian design and higher pension spending, and the income distribution moves spending little, under every variant. Whether it moves the design depends on the shape of the cost: little under the quadratic cost, which reproduces the weak relation between inequality and the design in figure 1.1, and under the linear-quadratic cost about as far as ageing, toward earnings-related benefits, with France's flatter voting patterns pulling as far the other way and prevailing when the two are combined, at 0.641 against the U.S. 0.738.

The paper belongs to the literature on politico-economic equilibria of social security

[2]When informal households can save, the state space includes two states.

[3]Appendix C.1 also considers a design chosen one period in advance and one chosen once and for all. At the U.S. calibration and without a cost of redistribution, each of the three timings gives a corner at every intertemporal elasticity of substitution the online appendix reports: the Beveridgean one, except that a design chosen once and for all is the Bismarckian one at high elasticities.

(e.g. Cooley and Soares 1999; Galasso 1999; Forni 2005; Gonzalez-Eiras and Niepelt 2008; Song 2011), surveyed by Galasso and Profeta (2002). As in Gonzalez-Eiras and Niepelt (2008) and Song (2011), under log-GHH preferences the equilibrium has no strategic effects and taxes increase with the political power of retirees, income inequality and ageing; as in Uchida and Ono (2024), the equilibrium tax is a function of parameters and demographics alone; and as in Gonzalez-Eiras and Niepelt (2005), the effect of ageing under CRRA preferences is close to the logarithmic one after recalibration. What we add is a pension system that differs in two dimensions, in coverage and in the earnings link, and workers who differ in productivity and in their propensity to vote. Neither adds a state when informal households are hand-to-mouth, and the calibrated model quantifies what each feature of the design does to the tax rate, labor supply and savings.

On the design of the system, Galasso and Profeta (2002) leave open why some systems are Beveridgean and others Bismarckian, and why the Bismarckian ones are larger, two questions that a vote on the size of the system alone cannot answer. Casamatta et al. (2000) study the political support of Bismarckian against Beveridgean benefits, Conde-Ruiz and Profeta (2007) the joint choice of size and design, Koethenbuerger et al. (2008) the trade-off by which more redistributive systems are smaller, and Cremer et al. (2007) the choice of type and generosity. We embed the choice in a dynamic equilibrium with capital accumulation and labor supply, in which the design is chosen with the tax at every date, and take it to the data.

Section 2 presents the model and the equilibrium concepts. Section 3 derives the equilibrium under log-GHH preferences, in which the tax is a function of parameters and demographics alone, and its comparative statics. Section 4 describes how the equilibria are computed outside that case. Section 5 calibrates the model to Argentina and compares its prediction with the observed reform, section 6 calibrates it to the U.S., France and the UK and ranks the determinants of pension spending, and section 7 lets the electorate choose the design. Section 8 concludes. The appendix collects proofs, derivations, and calibration details. An online appendix[4] collects auxiliary tables and figures and presents various robustness checks, and technical documentation accompanies the paper (Berg and Gonzalez-Eiras 2026).

2 The Model

2.1 Demographics, Inequality, and Institutions

We consider an economy populated by overlapping generations of workers and retirees. Within each generational cohort, we consider $J+1$ household types indexed $j = 0, \ldots, J$. The $j > 0$ household types allocate time between labor and leisure, pay taxes on labor

[4] championape.github.io/SSDclaude

income, consume, and save. When they reach retirement, they spend the return on their savings and any social security benefits they may receive, and die. Type $j = 0$ households similarly allocate time between labor and leisure, consume, and save for retirement. However, they supply labor to an informal labor market, thus not subject to labor income taxes, and they save for retirement using an informal savings technology. The distribution of type $j > 0$ households is given by γ_j with $\sum_{j>0} \gamma_j = 1$, and the share of type $j = 0$ households is γ_0. The ratio of workers to retirees in period t follows a deterministic process, and is given by ν_t.

The government runs a pay-as-you-go (PAYG) pension system with a balanced budget. Each period, the government imposes a tax on labor supplied in the formal sector and transfers the collected amount directly to retirees. We consider social security systems that differ along two dimensions: whether workers need to qualify to receive full benefits or not and the type of benefits provided, either tied to earnings or flat-rate. We describe the different systems in section 2.4.

Policy decisions are taken by a government that acts in the interest of voters, but lacks commitment.

2.2 Technology

In the formal sector, a continuum of competitive firms transform aggregate capital and labor into output. We define the aggregate formal labor supply as

$$H_t = N_t h_t, \qquad h_t \equiv \sum_{i>0} \gamma_i \eta_i h_{t,i}, \tag{1a}$$

where N_t is the number of formal workers at time t, $h_{t,i}$ is time allocated to labor by household type i, and η_i is their labor productivity, normalized such that $\sum_{i>0} \gamma_i \eta_i = 1$. Capital is owned by formal retirees and fully depreciates after a period. The economy-wide capital stock per formal worker, k_t, therefore corresponds to the economy-wide per capita savings of formal workers in the previous period, s_{t-1}, normalized by ν_t. We define the aggregate capital stock from

$$K_t = N_{t-1} s_{t-1}, \qquad s_{t-1} \equiv \sum_{i>0} \gamma_i s_{t-1,i}. \tag{1b}$$

The representative firm produces output using labor and capital and a conventional Cobb-Douglas technology

$$Y_t = K_t^{\alpha} H_t^{1-\alpha} \tag{2}$$

In equilibrium this implies the following factor prices:

$$R_t = \alpha \left(\frac{s_{t-1}}{\nu_t h_t}\right)^{\alpha-1}, \qquad w_t = (1-\alpha)\left(\frac{s_{t-1}}{\nu_t h_t}\right)^{\alpha}. \tag{3}$$

Informal households produce with a linear technology, so the informal wage w_t^0 and the return on informal savings R_t^0 are proportional to the formal sector's factor prices.

2.3 Preferences and Household Choices

Households value consumption and leisure, and discount the future at factor $\beta \in (0,1)$. Representing utility by function $u(c,h)$ with $u'_c \geq 0, u'_h \leq 0, u''_{cc} \leq 0, u''_{hh} \geq 0$, the welfare of a young household of type i born in period t is defined by

$$u\left(c^i_{1,t}, h_{t,i}\right) + \beta u\left(c^i_{2,t+1}, 0\right). \tag{4}$$

The budget constraints for formal and informal workers are given by:

$$c^i_{1,t} = w_t \eta_i h_{t,i}(1-\tau_t) - s^i_t, \qquad i > 0, \tag{5a}$$
$$c^0_{1,t} = w^0_t \eta_0 h_{t,0} - s^0_t, \tag{5b}$$

where τ_t is the social security tax rate. As retirees, households receive a pension b^i_{t+1} that may depend on the individual household's contribution, and thus on their labor supply, h^i_t:

$$c^i_{2,t+1} = s^i_t R_{t+1} + b^i_{t+1}(h_{t,i}), \qquad i > 0, \tag{5c}$$
$$c^0_{2,t+1} = s^0_t R^0_{t+1} + b^0_{t+1}(h_{t,0}). \tag{5d}$$

Optimization yields the first order conditions that represent a leisure/consumption tradeoff and a conventional Euler condition for intertemporal consumption smoothing:[5]

$$-\frac{\partial u_i}{\partial h_{t,i}} = \frac{\partial u_i}{\partial c^i_{1,t}}\left(w_t(1-\tau_t)\eta_i + \frac{\partial b^i_{t+1}/\partial h_{t,i}}{R_{t+1}}\right), \qquad i > 0 \tag{6a}$$
$$-\frac{\partial u_0}{\partial h_{t,0}} = \frac{\partial u_0}{\partial c^0_{1,t}}\left(w^0_t \eta_0 + \frac{\partial b^0_{t+1}/\partial h_{t,0}}{R^0_{t+1}}\right), \tag{6b}$$
$$\frac{\partial u_i}{\partial c^i_{1,t}} = \beta R_{t+1}\frac{\partial u_i}{\partial c^i_{2,t+1}}, \qquad i > 0 \tag{6c}$$
$$\frac{\partial u_0}{\partial c^0_{1,t}} = \beta R^0_{t+1}\frac{\partial u_0}{\partial c^0_{2,t+1}}. \tag{6d}$$

[5] We are using shorthand notation $u_i \equiv u(c_i, h_i)$ with timing implicit by the variable we take derivatives with respect to.

Optimal savings and labor supply decisions can, in general, be written as

$$h_t^i = h_t^i\left(s_{t-1}, s_t, h_t, h_{t+1}, \tau_t, \tau_{t+1}\right) \tag{7a}$$

$$s_t^i = s_t^i\left(s_{t-1}, s_t, h_t, h_{t+1}, \tau_t, \tau_{t+1}\right), \tag{7b}$$

where the dependence on h_{t+1}, τ_{t+1} comes from the pension system.

2.4 Social Security System

The government runs a pay-as-you-go pension system with a balanced budget. Every period, the government taxes labor income in the formal sector and pays pensions to current retirees.

We consider systems that differ along two dimensions. First, in *contributive* systems workers have to meet a contribution requirement to receive full pension benefits. In contrast, in a *universal* system all workers are eligible to receive pensions upon retirement, regardless of whether they contributed to it or not.

Second, pension schemes also differ according to the relationship between the level of contributions and benefits. *Bismarckian* systems provide earnings-related benefits; this gives workers an additional incentive to work as they perceive this increases their future pensions. In contrast, *Beveridgean* systems offer flat payments and lack this incentive. We model this by making benefits to consist of a common amount for all formal workers and a second component that is proportional to past labor income.[6]

Benefits are thus given by:

$$b_t^i = \left[\theta h_{t-1,i}\eta_i + (1-\theta)h_{t-1}\right]\bar{b}_t \tag{8a}$$

$$b_t^0 = \epsilon h_{t-1}\bar{b}_t \tag{8b}$$

where ϵ determines the level of universal pensions received by informal workers, θ measures how Bismarckian are the benefits of formal workers, and $\bar{b}_t$ is the average formal worker's pension benefits. With $\epsilon = 0$ the system is purely contributive and there are no pensions for informal workers. With $\epsilon = 1 - \theta$ they receive the basic common pension of formal workers, and with $\epsilon = 1 - \theta + \theta\eta_1 h_{t-1,1}/h_{t-1}$ the system is purely universal in the sense that informal workers receive the same pensions as the type of formal working household with lowest productivity.

The balanced budget requirement implies that

$$\nu_t w_t \tau_t h_t = \sum_{j\geq 0} \gamma_j b_t^j$$

[6]This formulation has been frequently used in the study of redistributive social security, see e.g. Casamatta et al. (2000).

This implies that the average pension rate $\overline{b}_t$ is:

$$\overline{b}_t = \frac{\nu_t w_t h_t \tau_t}{h_{t-1}(1+\gamma_0 \epsilon)}. \tag{8c}$$

2.5 Elections

Elections take place at the beginning of each period. We assume that preferences are aggregated through probabilistic voting.[7] Thus, policy maximizes a convex combination of the objective functions of all groups of voters, where the weights reflect the groups' sizes and their responsiveness to policy changes. We allow for age related variation in responsiveness, reflected in a per capita political influence weight of unity for young voters and a per capita weight of $\omega \geq 0$ for retired voters.

2.6 Equilibrium

Competitive Equilibrium. In economic equilibrium, firms and households optimize given factor prices, policies, and a vector of aggregate states z_t, which includes exogenous demographics as well as the distribution of states $\{s_{t-1,j}, h_{t-1,j}\}_j$. In a rational expectations equilibrium, the path of policies $\{\tau_s\}_{t \leq s \leq T}$ fully determines an allocation and price system.

Conditional on some initial states z_0 and a policy sequence $\{\tau_t\}_{0 \leq t \leq T}$, a competitive equilibrium is given by an allocation and price system such that

i. households optimize;

ii. aggregate states are defined according to (1), markets clear, and factor prices are determined according to (3) for all t; and

iii. the government budget constraints in (8) are satisfied for all t.

Politico-economic equilibrium. In politico-economic equilibrium political decision makers optimally choose tax rates, taking all implications of their actions into account and forming rational expectations about future policy choices. We assume that these choices are Markov, i.e. they are functions of the fundamental state variables. The decision maker at date t takes the distribution of states $\{s_{t-1,j}, h_{t-1,j}\}$ as well as the continuation policy $\tau_t^{+1}(z_t)$ as given. Furthermore, given the continuation tax function, the policymaker takes as given the following law of motion for the state variables as a function of current policy choices:

$$z_{t+1} = \zeta_t \left(z_t, \tau_t, \tau_t^{+1} \right). \tag{9}$$

[7] See Lindbeck and Weibull (1987).

Subject to (3), (8), (9), the policymaker chooses τ_t to maximize

$$\mathcal{W}_t\left(z_t, \tau_t; \tau_t^{+1}(z_{t+1})\right) \equiv \sum_j \gamma_j \mu_j \left[\omega \mathcal{O}_j(z_t, \tau_t) + \nu_t \mathcal{Y}_j\left(z_t, \tau_t; \tau_t^{+1}(z_{t+1})\right)\right],$$

where the objective function is the weighted sum of indirect utilities from retirees $\mathcal{O}_j$ and workers $\mathcal{Y}_j$, with weights reflecting relative population sizes through ν_t and γ_j, relative political weight of retirees ω, and propensity to vote μ_j.

Definition 1. *A* politico-economic equilibrium *as of period t conditional on z_t consists of a sequence of tax functions, $\{\tau_\iota(\cdot)\}_{\iota \geq t}$; a sequence of continuation tax functions, $\{\tau_\iota^{+1}(\cdot)\}_{\iota \geq t}$; a sequence of laws of motion for the state variables, $\{\zeta_\iota(\cdot)\}_{\iota \geq t}$; policy choices, $\{\tau_\iota^\star\}_{\iota \geq t}$; and a competitive equilibrium allocation such that*

i. tax functions are optimal subject to continuation tax functions:

$$\tau_\iota(z_\iota) \in \arg\max_{\tau_\iota} \mathcal{W}_\iota(z_\iota, \tau_\iota; \tau_\iota^{+1}(\cdot)) \text{ for all } z_\iota, \text{ s.t. } (3), (8), (9), \ \iota \geq t;$$

ii. continuation tax functions are consistent with tax functions:[8]

$$\tau_{\iota-1}^{+1}(z_\iota) = \tau_\iota\left(z_\iota, \tau_\iota^{+1}(z_{\iota+1}(\cdot))\right) \text{ for all } z_\iota, \iota \geq t;$$

iii. laws of motion are consistent with the policy and continuation policy functions according to (9);

iv. equilibrium tax choices are generated by the continuation tax function,

$$\{\tau_\iota^\star\}_{\iota \geq t} = \tau_{\iota-1}^{+1}(z_\iota),$$

and $\{\tau_\iota^\star\}_{\iota \geq t}$ implements a competitive equilibrium allocation.

For infinite horizon and a recursive time-autonomous structure, the policy and continuation policy functions are time-autonomous functions of the state as well, and conditions i. and ii. above are combined in a fixed point requirement.

3 Logarithmic-GHH Preferences

This section derives the model's analytical results. We follow Greenwood et al. (1988) and assume

$$u(c_{1,t}^i, h_{t,i}) = \ln\left(\tilde{c}_{1,t}^i\right), \qquad \tilde{c}_{1,t}^i \equiv c_{1,t}^i - \frac{X_i \xi}{1+\xi}\left(h_{t,i}\right)^{\frac{1+\xi}{\xi}} \tag{10}$$

[8] With a finite horizon, $\tau^{T+1}(z_T) = \emptyset$.

where ξ is the Frisch elasticity of labor supply and X_i is a type-specific taste for leisure. Proposition 1 characterizes the economic equilibrium under these log-GHH preferences. If informal households are in addition hand-to-mouth, the politico-economic equilibrium carries no state (proposition 2) and the tax rate is a function of parameters and demographics alone. The quantitative sections keep log-GHH preferences as the benchmark and let informal households save for retirement.

3.1 Economic Equilibrium

With log-GHH preferences, formal households i supply labor and save according to

$$h_{t,i} = \left(\frac{\eta_i w_t (1-\tau_t) + \frac{\theta \bar{b}_{t+1} \eta_i}{R_{t+1}}}{X_i} \right)^{\xi} \tag{11a}$$

$$s_{t,i} = \frac{\beta}{1+\beta} \left[w_t \left(1-\tau_t\right) h_{t,i} \eta_i - \frac{X_i \xi}{1+\xi} \left(h_{t,i}\right)^{\frac{1+\xi}{\xi}} \right] - \frac{b^i_{t+1}/R_{t+1}}{1+\beta}. \tag{11b}$$

Appendix A shows that in economic equilibrium every relevant variable x (savings, labor supply and consumption) is a power of capital per worker times a coefficient,

$$x_t = \Theta_{x,t} \left(\frac{s_{t-1}}{\nu_t} \right)^{\sigma_x},$$

where the coefficient $\Theta_{x,t}$ depends on parameters and taxes and the exponent σ_x on parameters only.

Proposition 1. *In economic equilibrium we have the following:*

i. Aggregate labor is decreasing in τ_t, ϵ and γ_0 and increasing in τ_{t+1}, θ. Aggregate savings are decreasing in τ_t, τ_{t+1}, and θ and are increasing in ϵ and γ_0.

ii. Labor income inequality and relative labor supply are independent of pension policy and given by

$$\frac{\eta_i h_{t,i}}{h_t} = \frac{\eta_i^{1+\xi}}{X_i^{\xi} \Gamma_h}, \qquad \Gamma_h \equiv \sum_i \frac{\gamma_i \eta_i^{1+\xi}}{X_i^{\xi}}. \tag{12}$$

iii. Savings of formal households relative to average savings are independent of τ_t and linear in τ_{t+1}:

$$\frac{s_{t,i}}{s_t} = \frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{1}{\Gamma_h} + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}}{1+\gamma_0 \epsilon} \frac{1-\theta}{1+\beta} \left(\frac{\eta_i^{1+\xi}}{X_i^{\xi} \Gamma_h} - 1 \right). \tag{13}$$

iv. When $\tau_{t+1} = 0$ or with fully Bismarckian pensions ($\theta = 1$), the heterogeneity in

savings exactly matches labor income inequality. An increase in τ_{t+1} or a decrease in θ amplifies the spread in private savings.

v. *Informal savings increase in τ_t and θ and decrease in τ_{t+1} and ϵ; current taxes also raise their ratio to formal savings.*

Part i of proposition 1 states that current taxes reduce labor supply and savings, while future taxes increase labor supply, since higher expected benefits make work more attractive, and reduce the need to save. A higher θ increases labor supply for the same reason; with GHH preferences this raises the marginal utility of $\tilde{c}_{1,t}$ and so, through the Euler equation, lowers aggregate savings. A higher ϵ shifts future benefits toward informal households and lowers the benefit formal households expect, so that they save more and work less, and a larger share of informal households has the same effects.

Parts ii and iii reduce the state space of the politico-economic equilibrium, as the next subsection shows. Part iv implies that higher future taxes or a lower θ, when pensions are redistributive, widen the dispersion of savings and thus of private wealth. Part v works through the universal pension, since informal earnings do not respond to pension policy. The pension's present value to an informal household is proportional to formal savings, with a factor that rises with future taxes and ϵ: current taxes and a higher θ, which lower formal savings, therefore raise informal savings, while future taxes and a higher ϵ raise the present value of the pension and lower them.

3.2 Politico-Economic Equilibrium

In a Markov equilibrium the distributions of past savings and past labor supply are in principle state variables of the political choice of the tax rate. Part ii of proposition 1 removes the distribution of labor supply, since taxes do not move relative labor supply.[9] If informal households save, their past savings are a further state, and one that both current and future taxes move. For the analytical results we therefore take informal households to be hand-to-mouth: they do not save, so that their Euler condition (6d) does not apply, and when old they receive an endowment tied to the informal wage, which can be read as work in old age at a lower productivity.[10]

With log-GHH preferences and hand-to-mouth informal households, the policymaker maximizes

$$\begin{aligned}\mathcal{W}_t(z_t, \tau_t; \tau_t^{+1}(z_{t+1})) = \sum_i \gamma_i \mu_i \left[\omega \ln\left(c_{2,t}^i\right) + \nu_t \left(\ln\left(\tilde{c}_{1,t}^i\right) + \beta \ln\left(c_{2,t+1}^i\right)\right)\right] \\ + \gamma_0 \mu_0 \left[\omega \ln\left(\tilde{c}_{2,t}^0\right) + \nu_t \left(\ln\left(\tilde{c}_{1,t}^0\right) + \beta \ln\left(c_{2,t+1}^0\right)\right)\right] \qquad (14)\end{aligned}$$

[9] This property of GHH preferences also holds under CRRA preferences (appendix A.1).

[10] We model old-age income as an endowment of goods to keep the model as symmetric with the general case as possible.

Proposition 2. *In politico-economic equilibrium with hand-to-mouth informal households, all consumption functions are log-separable in the endogenous state variables, making policy only a function of parameters, including system characteristics and demographics. Politico-economic equilibrium taxes are increasing in ω, decreasing in ν_t, and increasing in income inequality. Furthermore they are decreasing in θ. The effect of ϵ is ambiguous and depends on the pension's share of consumption of informal and formal retirees.*

The economic equilibrium is log-separable in the aggregate states (appendix A.2), and the one remaining vector of states, the distribution of formal households' savings $s_{t-1,i}/s_{t-1}$, is a function of parameters and the current tax alone; by the same token, the distribution carried into the next period does not depend on the current choice (appendix B.1). Despite two dimensions of pension design and heterogeneous households, the politico-economic equilibrium therefore shares the property of Gonzalez-Eiras and Niepelt (2008), Gonzalez-Eiras and Niepelt (2012), Song (2011), and Uchida and Ono (2024), among others, that policy is a function of parameters and demographics only, although the equilibrium conditions are more complex and in most cases can be solved only numerically.

Notwithstanding the orthogonality of current policy with state variables, the trade-offs underlying the equilibrium tax rates are dynamic in nature as they relate contemporaneous policy decisions to capital accumulation and thus future wages and tax revenue. The tractability of the model comes from specifying functional forms that render the factor price elasticities and the derivatives of the indirect utility functions orthogonal to the capital stock and relative labor supply and relative savings decisions independent of current taxes.

The comparative statics in ω and ν_t mirror those in Gonzalez-Eiras and Niepelt (2008) and Song (2011), and the latter also shows that the equilibrium tax rate increases with income inequality. A distinct contribution to the politico-economic literature on social security is the prediction that a more Bismarckian benefit formula weakens political support for taxation. The reason for this is that a higher θ shifts benefits from low-income to high-income retirees, and the former, with their higher marginal utility, weigh more in the political objective, so that retirees' support for taxes declines in the aggregate. A higher ϵ shifts benefits from formal to informal retirees. With log utility a retiree's support for the tax is the pension's share of their consumption, so the shift raises the support of informal retirees and lowers that of formal ones (appendix B.1 gives the exact condition for the sign). Taxes rise when informal retirees are poorer than formal ones and the pension is a comparable share of their consumption, as in Argentina (section 5), because their smaller pension then rises in a larger proportion than formal pensions fall.

When informal households save or preferences are CRRA, the tax rate depends on the state and the equilibrium has to be solved numerically; section 4 describes how.

4 Numerical Methods

This section describes how the politico-economic equilibria of sections 5–7 are computed, with the algorithms and the code in the technical documentation (Berg and Gonzalez-Eiras 2026). The quantitative politico-economic literature computes a time-invariant policy function (Krusell and Ríos-Rull 1999; Forni 2005; Song 2011) or compares steady states across demographic scenarios (Galasso 1999). Along a demographic transition, however, the Markov policy function is date-specific, since the policymaker at t faces its own mix of cohorts, ν_t, and a continuation policy set further along the transition. The distinction is immaterial in the equilibrium of proposition 2, which carries no state, but not under CRRA preferences or with informal savers, where the continuation reaches the tax through the state, or in the design choice, where a design chosen at t is weighed against the electorate of $t+1$.[11] We therefore compute date-specific policy functions by backward induction from a date T (appendix D), the end of a finite horizon or the start of a steady state in an infinite one. For a large T the two agree away from T. We report the finite-horizon solution, which approximates the Markov equilibrium in the limit of finite-horizon economies, the class in which Gonzalez-Eiras and Niepelt (2008) establish uniqueness.

In the general case with CRRA preferences, the distribution of labor supply remains independent of policy (part ii of proposition 1) and the distribution of past savings is pinned down by the current tax together with past aggregate savings (appendix A.1). While the distribution of past savings is indeed relevant for the political problem, we exploit that it only depends on the current tax and past aggregate savings (appendix A.1) to eliminate it from the state space, keeping only the level of past savings s_{t-1}: we work with the first order condition in τ_t derived with the distribution of past savings held fixed, and then replacing it with the one consistent distribution for each candidate tax and for each state s_{t-1}. Informal savers add the ratio $s_{t-1,0}/s_{t-1}$ of informal to formal savings to the state space, which also depends on τ_{t-1} (appendix B.2).

When the policy function features one or more states, the current tax moves the states and affects future policy choices. We measure the marginal effect by numerical differentiation of the $t+1$ solution grid, which the backward induction has already computed. A first order condition can have several roots, some of them minima, and the optimum can lie at a corner where it has none, so we evaluate the condition on a grid of taxes and test the corners and all its roots. A candidate passes if it maximizes the political objective with its own consistent distribution held fixed.[12]

[11] In our applications, the distinction has a modest effect on results: a time-invariant policy function, solved at each date's own ν_t and applied along the projected path, misses the tax rate by roughly 0.2 p.p., and by up to 0.5 p.p. in the Argentine calibration at the lowest intertemporal elasticity, and misplaces the design by up to 0.059.

[12] Several passing candidates would be ranked by payoff, an assumption about equilibrium selection

Each reported path starts from the steady state under the first period's policy. Parameters whose targets are equilibrium outcomes, such as the discount factor and the political weight of the old, are calibrated by re-solving the equilibrium path at each trial value until the targets of sections 5 and 6 hold in the calibration year.

Under log preferences the choice of the design in section 7 adds no state either. The design chosen at t applies from $t+1$ and depends neither on the design chosen in the previous period, which sets current benefits, nor on aggregate savings (appendix C.4), and the tax at each candidate design is found as above, without a state. Each date's choice is therefore a search over a grid of designs. Under CRRA preferences neither property survives: the equilibrium is a backward recursion over policy functions of the state (s_{t-1}, θ_t).

5 Pension System Reform in Argentina

Argentina is a middle-income country with high labor informality. In 1994 the social security system was reformed tying full benefits to 30 years of contributions, which left about a third of the elderly without a pension by 2004.[13] From 2005 a series of tax amnesties let workers with shorter contribution records draw a pension. They added 2.7 million beneficiaries by 2011 (Rofman and Apella 2015), raising the number of pensioners from about 3 to 5 million (figure 5.1) and the coverage of the elderly from 68% to over 90% (Rofman and Oliveri 2012). Thus, in a few years, Argentina experienced a significant increase in non-contributive pensions: Between 2005 and 2010 social security spending increased by 1.9% of GDP with most of this increase, 1.8% of GDP, due to the tax amnesties (Cetrángolo and Grushka 2020).

The following outlines the calibration procedure and presents the main results from the pension reform. Appendix E keeps calibration details and the online appendix provides supplementary evidence from the reform.[14]

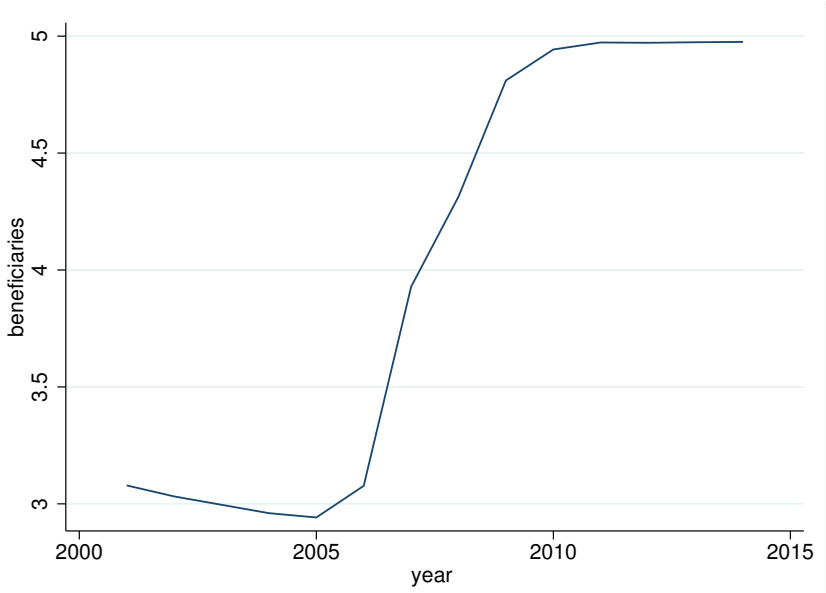
Calibration. We calibrate the model with informal savers to Argentina before the reforms and read the tax amnesties as an increase in ϵ. Informal households work when young and save for retirement, facing the same rate of return on savings as formal households. A model period is thirty years, the calibration year is 2010, and every household has the same propensity to vote. Table 1 reports the result. The pre-reform $\epsilon = 0.21$ is the basic pension of retirees without a full contribution record, 70% of the benefit of the

that none of our reported equilibria requires: exactly one passes at every state and date of every solve that records the count (online appendix).

[13] The reform also let workers opt for a fully funded system, whose funds were nationalized in December 2008.

[14] Appendix E holds the data, the identifying formulas and the sources of the calibration. The online appendix has additional evidence including how the design (ϵ, θ) shapes equilibrium outcomes and how different calibrations affect results.

Figure 5.1: Pension coverage in Argentina



Note: Beneficiaries of the national pension system (SIPA) and of non-contributive old-age pensions, annual averages in millions. Source: Cetrángolo and Grushka (2020, table 6), from the Boletín Estadístico de la Seguridad Social.

least productive formal workers, in units of the average formal benefit and discounted because it starts five to ten years after the formal retirement age, and $\gamma_0 = 0.47$, informal households per formal household, puts informal households at 32% of all, the share of the population over 65 without a pension before the reform (Rofman and Oliveri 2012). The Bismarckian index $\theta = 0.84$ matches the observed ratio of replacement rates at mean and half-mean income. The reform raises ϵ to the universal level $\epsilon^U = 1 - \theta + \theta\eta_1 h_{t-1,1}/h_{t-1}$ of section 2, at which informal retirees receive the benefit of the least productive formal ones. We impose a formal-sector capital share of $\alpha = 0.35$, between the values implied by the survey's informal earnings and by shadow-economy estimates of informal output, a Frisch elasticity of $\xi = 0.30$, and a ratio of workers to retirees ν_t from thirty-year gross population growth in census data and projections, as in Gonzalez-Eiras and Niepelt (2008).

The remaining parameters are identified jointly with the equilibrium path. The formal productivities η_i match the income distribution across the four formal quartiles of the survey, with a taste for leisure X common to them whose level is pinned by the observed formal workweek of 42.5 hours.[15] Relative hours across quartiles are then a prediction, which reproduces the survey's ranking and overstates the hours of the top quartile, 1.18 against 1.10 times the formal average. Four parameters are solved against four targets in 2010: β against a capital-output ratio of 3.23, ω against an equivalent social security tax

[15]Identifying one X_i per quartile from the survey's relative hours instead makes hours data and leaves their average level unidentified (appendix E.5). Since X enters no aggregate, only η_i and X_i differ, and every result of this section is the same under both.

rate of 10.9%, which is pension spending of 7.1% of GDP before the 2001 crisis divided by the labor share of formal output, and η_0 and X_0 against the informal households' income and hours relative to the formal average.

Table 1: Calibration, Argentina

Parameter	**Value**	**Target**
ϵ (*pre*-reform)	0.21	Basic pension, 70% of the minimum, discounted
θ	0.84	Replacement rate dispersion
α	0.35	Factor income shares
ν_t	$[1.65, 0.96]$	30-year gross population growth rates
ξ	0.30	Elasticity of labor supply
β	0.65	Capital–output ratio of 3.23
X	1.94	Average formal workweek of 42.5 hours
η_i	$[0.64, 0.90, 1.14, 1.87]$	Income distribution
γ_0	0.47	Elderly without a pension, 32% of all households
ω	1.45	Social security tax of 10.9%
η_0	0.344	Informal relative income, eq. (E.1)
X_0	0.804	Informal relative working hours

Note: Our default specification relies on $\rho = 1$.

The reform. Table 2 reports the effect in 2010 of an unanticipated and permanent rise of ϵ to ϵ^U under log-GHH preferences, first with the tax rate held at its pre-reform path and then chosen politically. We find that taxes increase, which drives down savings and labor supply. With taxes held fixed, formal households save more and work marginally less, since the reform lowers the benefit they expect (proposition 1); informal households, who now hold a claim on the pension system, save less.

In 2010 the model predicts that the social security tax rate rises by 2.1 p.p., which at the labor share of formal output is pension spending of 1.4% of GDP, about three quarters of the 1.8% of GDP that the amnesties added. The reason for this is that the reform moves benefits toward informal retirees, who are poorer than formal ones and value benefits more at the margin.[16] The savings rate falls by 0.4 p.p. of GDP and the average workweek by a quarter of an hour, the fall in savings in line with the estimate of Gonzalez-Rozada and Ruffo (2016) for the amnesty. The tax response builds over time, to 3.5 p.p. above the pre-reform path by 2040, because informal households, now covered, save less for retirement and as retirees demand more taxation.

[16]The tax also rises when informal households are hand-to-mouth, consistent with condition (B.5) in appendix B.1.

Table 2: Pension system reform, year 2010

Scenario	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	10.92%	14.40%	42.54
Economic Equilibrium	10.92%	14.73%	42.46
Full effect	13.00%	13.97%	42.29

Note: The reform permanently shifts $\epsilon = 1 - \theta + \theta\eta_1 h_1/h$ at the calibration year, unanticipated. The "Economic Equilibrium" scenario solves for the new economic equilibrium path with taxes as in the Baseline scenario, but with the new permanent ϵ; the "Full effect" scenario lets taxes be determined by the politico-economic equilibrium. $\rho = 1$. The savings rate is savings relative to GDP.

The effect of the intertemporal elasticity of substitution. Since the environment has no uncertainty, we read the parameter ρ of CRRA preferences as the intertemporal elasticity of substitution (IES). The evidence on its size is inconclusive,[17] and we keep logarithmic utility as the benchmark. Figure 5.2 repeats the reform with the pre-reform economy calibrated at each ρ and traces the effects in 2010 and 2040 for taxes, savings rates, average workweek, and informal savings.

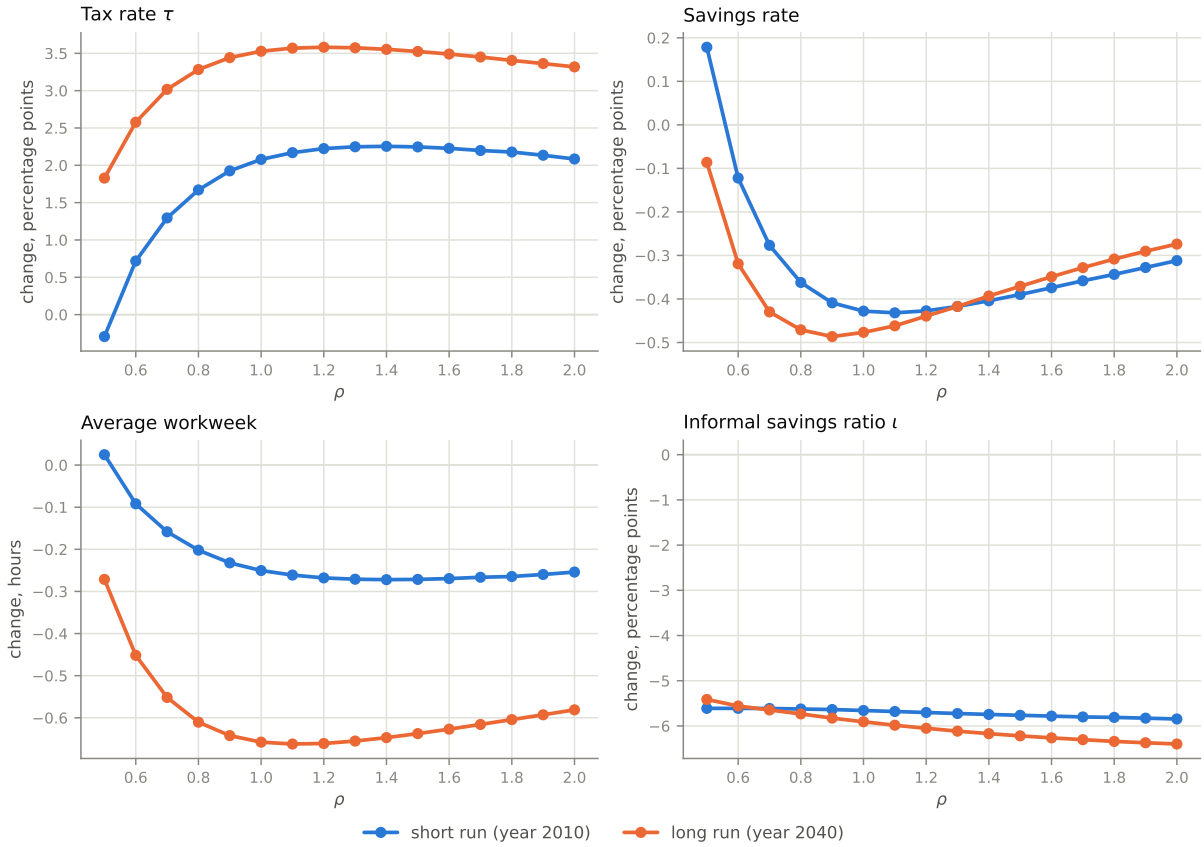
The effects of the pension reform are nonlinear in ρ, in particular for values lower than one. When $\rho < 1$ the effects of the pension reform are dampened across all metrics, and at the lowest values of the IES the tax rate even falls slightly on impact. The reason for this is that with a low IES, the young resist taxation more strongly, so the recalibrated economy needs a larger political weight of the old to reproduce the observed tax rate and responds less to a change in benefits. For an IES between 1 and 2 the results stay close to the log benchmark: the tax rises by 2.1 to 2.3 p.p. in 2010, the workweek shortens by roughly 0.25 hours in 2010, and the fall in the savings rate shrinks from 0.4 to 0.3 p.p. of GDP. The informal savings ratio falls by 5.7 to 5.8 p.p. throughout this range. Results are similar in 2040, albeit with tax and workweek effects amplified. Since Gonzalez-Rozada and Ruffo (2016) find that the amnesty reduced savings, we regard $\rho \in [1, 2]$ as the more plausible range.

6 Social Security in Rich OECD Countries

In order to study the determinants of pensions, we now calibrate the model to three rich OECD economies that differ markedly in pension spending, demography, income inequality, voting patterns and pension design: the U.S., France and the UK. In these economies nearly every worker qualifies for a contributive pension, so we set informality aside and

[17] Estimates from aggregate consumption data are low, which Guvenen (2006) reconciles with higher values from output and wealth data through limited asset market participation, and asset-pricing models with Epstein–Zin preferences and long-run or disaster risk require an IES above one (Bansal and Yaron 2004; Barro 2009).

Figure 5.2: Short- and long-run effects of the pension reform as a function of ρ



Note: Each panel is the change against the pre-reform path of the same ρ. The savings rate is savings relative to GDP, and the informal savings ratio ι is the savings of an informal household relative to the average savings of formal households.

the design reduces to one dimension, the extent to which benefits are earnings-related.[18] To explore what characteristics shape pensions, savings rates, and labor supply, we simulate counterfactual economies that are calibrated to the U.S. with one characteristic changed at a time.

As for the Argentina case, the following outlines the main results. Appendix F and the online appendix contain more details on the calibration and supplementary evidence.[19]

Calibration. We calibrate the three economies to 2020 in the same way as Argentina in section 5, with a few exceptions. First, with informality set aside, the population is split into three formal income groups. The first two groups are cut, as for Argentina, to have mean incomes of half the mean and the mean, the incomes at which the OECD's replacement rates identify θ. France's system is fully Bismarckian, $\theta = 1$ at any cuts, so

[18] Whitman et al. (2011) estimate that about 4% of individuals aged 62–84 in the U.S. in 2010 would never collect Social Security, most of them for lack of the 40 quarters of covered earnings needed to qualify.

[19] The online appendix holds every table behind the figures of this section, the same counterfactuals with the UK as the host economy, and the calibrations under the alternative treatment of the taste for leisure and across the intertemporal elasticity of substitution.

we group French households at the U.S. income percentiles instead, which puts the two countries' household types in one-to-one correspondence for the counterfactuals below. Second, the discount factor β matches the U.S. thirty-year interest rate rather than a capital–output ratio and is imposed on France and the UK. Third, where the Argentine calibration gives every household the same propensity to vote, the political objective here weights each income group by its observed propensity. In the U.S. the poor vote much less than the rich, while in France the propensity to vote is nearly flat across income groups. Fourth, since the model is one of intergenerational transfers, we add Medicare to U.S. pensions, in spending and in the calibration of θ, where as a benefit unrelated to income it makes transfers to the old more Beveridgean than pensions alone.

Table 3 reports the parameters. Population growth is highest in the U.S. and lowest in France, $\nu_{2020} = 1.34$ against 0.97, and the ratio of high to low productivity is about a third higher in the U.S. than in France, 3.08 against 2.29. The average taste for leisure is roughly twice the U.S. level in the UK and France. The calibrated political weight of the old is nearly the same in the U.S. and France, 1.45 and 1.42, and lower in the UK, 1.16.

Table 3: Calibration, the U.S., the UK and France

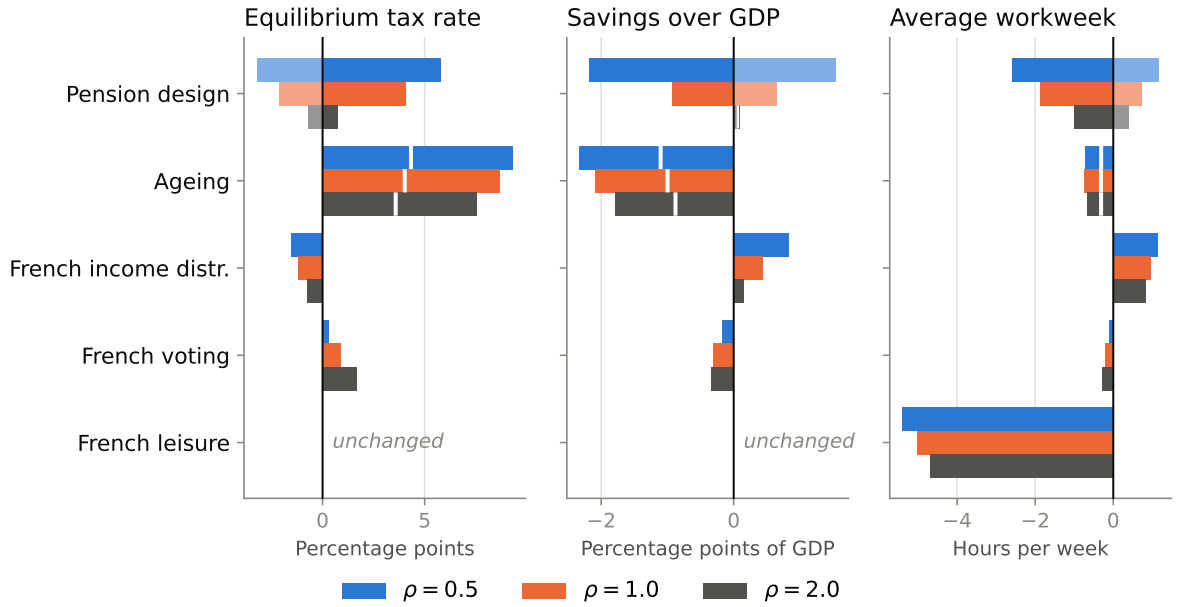
Parameter	**U.S.**	**UK**	**France**	**Identified by**
θ	0.74	0.56	1.00	Replacement rate dispersion
ω	1.45	1.16	1.42	$\tau^{US} = 14.4\%$, $\tau^{UK} = 11.9\%$, $\tau^{FR} = 21.3\%$
X	4.03	8.26	7.27	Average workweek
ν_{2020}	1.34	1.17	0.97	30-year gross population growth
η_H/η_L	3.08	3.06	2.29	Relative productivity, high to low income
β		0.76		30-year interest rate (U.S.)
ρ		1		Log-GHH preferences
ξ		0.30		Frisch elasticity of labor supply
α		0.30		Capital income share

Note: X is the population-weighted mean of X_i; its level is the hours unit, pinned for France and the UK by targeting average hours relative to the U.S. rather than in levels. The lower panel is common to the three countries; β is calibrated for the U.S. and imposed on the other two.

Determinants of pensions, savings, and labor supply. Figure 6.1 summarizes the counterfactuals. Each row changes one characteristic of the U.S. calibration: its pension design, population growth, income distribution, leisure preferences or voting patterns. Each panel reports the effect on the equilibrium tax rate, the savings rate and

the workweek in 2020, at the three values of the intertemporal elasticity ρ considered below. The counterfactuals are guided by France and the UK, particularly France, whose households can be grouped as in the U.S. and whose calibrated ω is close to the U.S. value. Each is a separate equilibrium path on which the changed characteristic holds throughout, so a row describes an economy that has always had that characteristic rather than the U.S. surprised in 2020.

Figure 6.1: The effect of various counterfactuals in the U.S.



Note: Full tables with results are printed in the online appendix. In the pension-design row each bar shows both corners, measured from the baseline: $\theta = 0$ in full colour and $\theta = 1$ in the light tone. In the ageing row the bar is acute ageing and the white line across it mild ageing. The savings rate is savings relative to GDP.

The tax panel identifies the pension design and ageing as the main determinants of pension spending. Going from a fully Bismarckian system to a fully Beveridgean one raises the equilibrium tax rate by 6.2 p.p. at $\rho = 1$ and lowers the savings rate by 1.6 p.p. of GDP. The reason for this is that of proposition 2: flatter benefits direct more of the revenue to poorer retirees, whose higher marginal utility gives them more weight in the political objective, so retirees support a larger system. Holding taxes fixed, an increase in θ would lower savings, but the endogenous fall in taxes more than offsets this and the net effect is positive. For labor supply, the economic-equilibrium and political effects of an increase in θ are aligned, and the tax response contributes as much as the direct effect or more.

Population ageing is the determinant that previous studies of the politico-economic equilibrium of pension spending have highlighted: Gonzalez-Eiras and Niepelt (2008) predict an increase in U.S. spending of around 3.6 p.p. towards the middle of this century because of the slowdown in population growth. In the data, including projections, ageing

is more acute in France than in the UK, and much more than in the U.S. We consider two scenarios, which roughly correspond to the data for the two European countries. In the first, *mild ageing*, we halve current and projected U.S. population growth; in the second, *acute ageing*, the population is constant, $\nu_t = 1$ for all t. Acute ageing raises the tax rate by 8.7 p.p. and mild ageing by 4.0 p.p.; the mild scenario is close to the demographic projection of Gonzalez-Eiras and Niepelt (2008).[20] When acute ageing drives up pension contributions significantly, this crowds out private savings by 2.1 p.p. of GDP at $\rho = 1$. Working hours fall by roughly 0.7 hours a week: the higher tax rate lowers the effective wage rate net of taxes, but this is partly offset by a capital deepening effect from ageing on wages. Thus, the net result is a modest drop in hours worked.

The characteristics that set France's households apart move the tax rate much less. At $\rho = 1$, France's flatter income distribution, imposed with θ held at its U.S. value, lowers it by 1.2 p.p.,[21] and its flatter voting profile, under which the poor vote about as often as the rich, raises it by 0.9 p.p. Rescaling the taste for leisure to France's level shortens the workweek by more than the whole gap to France's and leaves every rate at the baseline, since a proportional change in X_i scales hours, output and savings together. With the UK as the host economy the effects are smaller still, and France's income distribution raises the UK's tax rate, since at the UK's income cuts France is the more unequal of the two countries (appendix F.4).

The workweek panel ranks the determinants differently. France's leisure preferences move hours most and leave every rate unchanged. Among the characteristics that move the tax rate, the pension design moves hours most, by about two hours a week at the Beveridgean corner at $\rho = 1$, while ageing, the largest effect on taxes, moves them less than France's income distribution does at every ρ. The determinants of pension spending are thus not the determinants of hours worked.

The effect of the intertemporal elasticity of substitution. In general, our results above are robust across different ρ and the overall conclusion about the determinants of pension spending, savings, and labor supply holds. Effects are generally slightly amplified for lower values of ρ and dampened for higher values of ρ; the one exception where this pattern is reversed is the French voting profile. The IES plays a key role in the effect of the pension design: going from $\theta = 1$ to $\theta = 0$ raises the tax rate by 9.0 p.p. at $\rho = 0.5$ and by only 1.5 p.p. at $\rho = 2$. The reason for this is that a higher IES lowers the young's resistance to taxation, so the calibration needs a lower political weight of the old to match the observed tax rate, and the difference in benefits between rich and poor

[20] In Gonzalez-Eiras and Niepelt (2008), $\nu_{2000} = 1.38$ and the projected $\nu_{2080} = 1.23$, a fall of about 0.15, similar to the fall of 0.17 in 2020 in the mild ageing scenario.

[21] We read θ off the ratio of replacement rates at mean and half-mean income, so under France's flatter income distribution the same observed ratio would imply a considerably less Bismarckian system. Re-estimating θ would bundle a change in the design into a counterfactual about inequality.

retirees, through which θ moves the tax rate, counts for less; the savings rate and the workweek follow.

The calibrated parameters absorb institutions that the model does not contain. The political weight of the old, ω, stands for the way legislation is introduced, debated and passed, and for constitutional arrangements such as whether the political system is presidential or parliamentary. The taste for leisure, X, stands for preferences and also for institutions that shape labor supply, such as unionization, collective bargaining, labor-market regulation and statutory holidays, so the large differences in X across the three countries include these institutional differences.

We now return to figure 1.1, in which pension spending and the Bismarckian index both rise as population growth falls, and neither is related to income inequality. With the design given, the model accounts for the pattern in pension spending: ageing is the main determinant of the equilibrium tax rate, and the income distribution moves it several times less.[22] The relation between the design and population growth suggests that countries respond to ageing by tying benefits more closely to earnings, which a model that holds θ fixed cannot address. Section 7 takes it up by letting the electorate choose θ as well as τ.

7 Endogenous Pension Design

So far the Bismarckian index θ is calibrated while the tax rate is chosen politically. This section lets the electorate choose the design with the same objective $\mathcal{W}_t$, in the model without informal households, so one political process delivers the size of the system and its shape. The derivations are in appendix C, further results in appendix G, and the algorithms in the technical documentation (Berg and Gonzalez-Eiras 2026).

7.1 The deadweight cost of redistribution

Without a deadweight cost of redistribution the choice is a corner. An electorate that weighs retirees by their marginal utility wants all of the transfer the flat component performs: if propensities to vote do not rise with income, the design chosen with the tax is the Beveridgean corner, $\theta_t = 0$ (proposition 3 in appendix C.1). Propensities to vote that rise with income, as in the U.S., could overturn it, but at the U.S. calibration every timing of the choice, with the tax, one period in advance or once and for all, gives a corner (appendix C.1). The model lacks a mechanism that makes flat benefits costlier than earnings-related ones, and an interior design requires one.

[22] On the disposable-income Gini of figure 1.1, the U.S., at 0.42, is the third most unequal of the 29 countries, after Mexico and Turkey, and France, at 0.32, is the median.

The flat component redistributes within the retired cohort, and the transfer it performs is an implicit tax. A retiree of type i receives $\Pi_t \tau_t \left[\theta_t + (1-\theta_t)/y_i\right]$ per unit of past earnings, where y_i is the labor income of type i relative to the mean and $\Pi_t \equiv \nu_t w_t h_t/(w_{t-1}h_{t-1})$ the pay-as-you-go return factor, against $\Pi_t \tau_t$ from an earnings-related system of the same revenue. The flat component therefore levies $t_i = \Pi_t \tau_t (1-\theta_t)\,(1 - 1/y_i)$ per unit of past earnings, equation (C.4) in appendix C.2: a tax on retirees whose past earnings were above the mean, a subsidy to those below it, and a pure transfer in the aggregate. It is the tax component of the contribution in the sense of Summers (1989), the part of the payroll charge that buys the contributor no benefit. The model already contains one response to it. The flat benefit depends on aggregate hours and not on own hours, so the marginal wedge of the flat component is the same for every type, and hours respond to it with the Frisch elasticity through the labor supply function (A.6). What the model lacks is an extensive margin, whether to participate, to work formally or to report earnings, on which a worker compares the earnings at stake with what they buy, so that the average wedge t_i is what matters, and it differs across types.

What such a wedge costs is the subject of a large literature. A tax costs resources beyond the revenue it raises, by a loss that is quadratic in the wedge (Harberger 1964), and redistribution carries the same loss in Okun's image of the leaky bucket, in which part of what is carried from the rich to the poor does not arrive (Okun 1975). The tax component of pension contributions acts as a tax on the extensive margin: Disney (2004) splits contributions in OECD countries into a tax and a saving component and finds that the tax component lowers the participation of women while the saving component raises it, and Kumler et al. (2020) find that the underreporting of wages to Mexico's social security agency fell for the workers whose benefits a reform tied more closely to reported wages. On that margin the relevant wedge is the average tax on the earnings at stake rather than the marginal rate, and the cost of taxation is larger than its hours response alone would suggest (Saez 2002; Kleven and Kreiner 2006); the participation and taxable-income elasticities that measure it are of the order of 0.1 to 0.4 (Chetty et al. 2011; Saez et al. 2012). Finally, the size of a system is related to its design: Conde-Ruiz and Profeta (2007) document that more redistributive systems are smaller, and Koethenbuerger et al. (2008) account for it by an efficiency–redistribution trade-off in which flat benefits carry a larger efficiency cost per unit of size.

We take the ingredients of the cost from this literature and let it scale the revenue that reaches retirees, in the sense of Okun's bucket: part of what the flat component carries does not arrive. Per unit of mean past earnings the flat component carries $\frac{1}{2}\Pi_t \tau_t (1-\theta_t) M$, with $M \equiv \sum_i \gamma_i |y_i - 1|$ the mean absolute deviation of relative income, since what is taken above mean income is what arrives below it (appendix C.2). The leak has two parts. A constant share of every unit carried is Okun's leak proper, the administrative and political cost of the transfer rather than a behavioral response; it costs a share of

revenue proportional to $(1-\theta_t)M$ and prices the first unit of redistribution. A share that rises with the load is the Harberger loss of the implicit taxes on the margins the model lacks: quadratic in each type's wedge and summed over types, it is a share proportional to $\tau_t(1-\theta_t)^2\tilde{V}$, with $\tilde{V} \equiv \sum_i \gamma_i(y_i-1)^2/y_i$ the dispersion of relative incomes, in which a given transfer counts for more against a smaller income, and it leaves the first unit of redistribution free at the margin. We scale both parts by the size of the system, so that a larger system is costlier to keep redistributive, the trade-off by which Koethenbuerger et al. (2008) account for the finding of Conde-Ruiz and Profeta (2007), read from the other side. Taken out of revenue, the budget is

$$\nu_t w_t \tau_t h_t \, f(\theta_t, \tau_t) = \sum_i \gamma_i b_t^i, \tag{15}$$
$$f(\theta_t, \tau_t) = \exp\Big\{ -\tau_t(1-\theta_t)\Big[\lambda_1 M + \tfrac{1}{2}\lambda_2 \tilde{V}(1-\theta_t)\Big]\Big\},$$

where $1-f$ is the share of revenue lost, to first order $\lambda_1\tau_t M(1-\theta_t) + \frac{1}{2}\lambda_2\tau_t\tilde{V}(1-\theta_t)^2$, and $\lambda_1, \lambda_2 \geq 0$ are the cost parameters (appendix C.2 derives the form from the implicit tax).[23] We compare three variants. The quadratic cost, $\lambda_1 = 0$, is the Harberger loss alone; the linear cost, $\lambda_2 = 0$, is Okun's leak alone; the linear-quadratic cost has both, written as a scale and a shape, $\lambda_1 = \lambda\kappa$ and $\lambda_2 = \lambda$, and is our preferred specification.

Three properties hold for every variant. Nothing is lost when nothing is redistributed: $f = 1$ at $\theta_t = 1$ and, for every design, when all incomes are equal, where M and $\tilde{V}$ vanish together. The share lost rises with the size of the system, through the factor τ_t. And the marginal cost of the first unit of redistribution is $\lambda_1\tau_t M$: zero under the quadratic cost, under which no electorate chooses the Bismarckian corner, and positive under the two with a linear term, under which an electorate chooses the corner once λ_1 exceeds a threshold. The threshold is proportional to the inverse of the size of the system, since the young's stake in the design and the weight of the pension in their lifetime resources are both proportional to the tax (the technical documentation derives it), and it rises with a flatter voting profile and falls with ageing (appendix C.2); the design objective is single-peaked at the three economies we calibrate, so this condition at the corner decides the choice. The solution of sections 2 to 4 carries over with $f\theta_t$ and $f(1-\theta_t)$ in place of θ_t and $1-\theta_t$ (appendix C.3); the cost cancels from the replacement-rate ratio, so $\theta^* = 0.738$ is unchanged under every variant, and adds one factor to the tax condition, $1 + \ln f$, as a larger system is worth less to a retiree at the margin than the revenue it collects. The design is chosen one period in advance, the Markov timing in which it moves labor

[23] The exponential form keeps f positive at every size of the system and nests the model without the cost at $\lambda_1 = \lambda_2 = 0$. The factor τ_t on the linear term is a choice: a constant share of the amount carried is, as a share of revenue, independent of the size of the system, and we let it rise with the size as the quadratic share does, so that one factor serves every variant in the tax condition and a larger system lowers the cost parameter at which an electorate chooses the Bismarckian corner (appendix C.2).

supply and savings before it binds and the way benefit formulas are legislated; under log preferences the choice depends on neither the inherited design nor savings (appendix C.4). With the cost, a more Bismarckian design also raises the total the young will receive, which makes an interior choice possible; the variants differ in how fast the marginal cost falls as the design becomes more earnings-related, and so in how far a characteristic moves the chosen design.

The cost stands for the margins the model lacks, participation, formality and the reporting of earnings, and for the administrative and political costs of redistribution; its parameters are calibrated rather than derived. The quadratic one is larger than estimated elasticities would give, and the linear one has no elasticity benchmark (appendix C.2); their discipline is the cross-country test of section 7.2.

7.2 Calibration and a cross-country test

Table 4 reports the three calibrations. Under the quadratic and the linear cost one parameter is calibrated so that θ_{2020}, the design chosen in 1990 on the equilibrium path with the choice binding from the first period, equals θ^*, with (β, ω) recalibrated at every trial value. Under the linear-quadratic cost the scale and the shape are pinned jointly by the U.S. and the UK designs, the UK at its own income cuts as in section 6, so that both electorates re-elect their observed designs under one pair (λ, κ). The calibration pins 2020 only; every later design is a prediction. The linear-quadratic cost has $\lambda = 5.237$ and $\kappa = 0.129$, so a linear parameter of 0.674; the linear cost has $\lambda_1 = 1.710$ and the quadratic $\lambda_2 = 8.643$. The share of revenue the observed U.S. design loses in 2020 rises with the linear term, 1.8%, 2.6% and 3.6% under the quadratic, the linear-quadratic and the linear cost, while the share a flat benefit of the same size would lose falls, 23.8%, 19.8% and 13.2% (table 4). One datum pins each parameter, so the test of a specification rests on the economies not used: the UK and France under the one-parameter costs, France under the linear-quadratic cost.

Under the U.S. parameters the UK (observed design 0.560, tax rate 11.9%) chooses 0.625 under the quadratic cost and the Beveridgean corner under the linear one, while France (1.000, 21.3%) chooses 0.749 and the Bismarckian corner. Under the linear-quadratic cost, with the UK on its design by construction, France is predicted at 0.806: interior, further above the U.S. design than under the quadratic cost, but short of its corner. The ordering of the three designs is the observed one under every variant.

The own parameters say what each design asks of the cost (table 10 in appendix G.1). The UK's design pins a small linear term, under which France is predicted interior, and France's corner takes a linear term about twice as large, under which the UK is predicted far below its design; no pair of parameters that keeps the U.S. on its design reconciles the two (appendix G.1). What the second parameter buys is a corner within reach at a finite

parameter and a response to characteristics that the curvature alone damps; what the test asks of it is one ordering, and its other side, the shape France's corner pins with the UK as the prediction, misses the UK's design by a similar margin, 0.397 against 0.560. The linear cost reaches France's corner at the U.S. parameter, but it is knife-edge: with no curvature in the leak, the chosen design responds to a characteristic as a switch, and the UK's electorate falls to the Beveridgean corner. What separates the UK from France is the size of the system, tax rates of 11.9% against 21.3% (table 3), as in the cross-section of figure 1.1, and the specification builds this reading in by scaling the linear term with the tax rate: the two have nearly the same dispersion of relative incomes, 0.223 and 0.218, about half the U.S. 0.436, and a larger system lowers the threshold at which an electorate chooses the corner in proportion (appendix C.2); under the linear-quadratic cost the amount France's flat component carries, $M = 0.422$ against the UK's 0.292, adds to it. The UK regrouped at the U.S. income percentiles and the calibrations at $\rho \neq 1$ are in appendices G.1 and G.3.

Table 4: Three specifications of the cost of redistribution

	U.S.				**θ chosen**		**Own value**	
Cost	λ_1	λ_2	$f(\theta^*)$	$f(0)$	**UK**	**France**	**UK**	**France**
Linear-quadratic	0.674	5.237	0.974	0.802	0.560	0.806	5.237	≥ 9.793
Linear	1.710	0	0.964	0.868	0.000	1.000	2.414	≥ 1.312
Quadratic	0	8.643	0.982	0.762	0.625	0.749	7.264	–

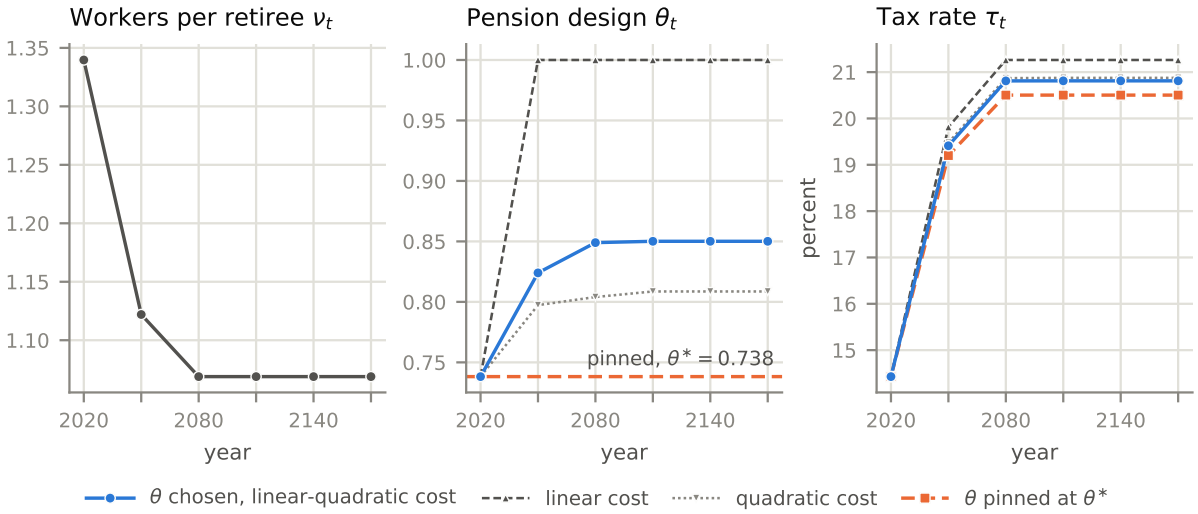
Note: Log preferences, $\rho = 1$. The cost is that of equation (C.8), $\ln f(\theta_t, \tau_t) = -\tau_t(1-\theta_t)\big[\lambda_1 M + \frac{1}{2}\lambda_2 \tilde{V}(1-\theta_t)\big]$, with $M = \sum_i \gamma_i |y_i - 1|$ the amount the flat component carries and $\tilde{V}$ the dispersion of relative incomes; the linear cost sets $\lambda_2 = 0$ and the quadratic $\lambda_1 = 0$. The linear and the quadratic costs have one parameter each, calibrated so that the U.S. electorate re-elects its observed design θ^* with (β, ω) recalibrated at each trial value; the linear-quadratic cost has two, pinned jointly by the U.S. and the UK designs at $\kappa = \lambda_1/\lambda_2 = 0.129$, so that the UK's own value is the U.S. one and France's design is a prediction. $f(\theta^*)$ and $f(0)$ are the shares of revenue that reach U.S. households at the observed design and at a flat benefit, at the 2020 tax rate. "θ chosen" is the design in 2020 on the economy's own freely simulated path under the U.S. parameters; "own value" is the value of the parameter the specification calibrates (λ_1 under the linear cost, λ_2 under the others, at the U.S. κ under the linear-quadratic) at which that path re-elects the observed design, with β imposed from the U.S. and ω recalibrated at each trial value. France's observed design is the corner $\theta = 1$, so its own value is the smallest at which its electorate chooses the corner, a bound ($\geq$); under the quadratic cost no finite value reaches it (–). The observed designs and 2020 tax rates are those of table 3 and the dispersions of each economy are in table 10; the UK is at its own income percentiles, and at the U.S. ones in the same table.

7.3 Ageing and inequality

We read three of the counterfactuals of section 6 twice (table 5), with the design pinned at the U.S. value and chosen, both under the cost, so their distance is the political response, one block per variant. On the baseline path (figure 7.1) the chosen design rises with ageing under every variant. Under the linear-quadratic cost it rises from 0.738 in 2020 to

0.824 in 2050 and 0.850 from 2110 on, tracking the fall in workers per retiree from 1.34 to 1.07, and the tax rate rises from 14.4% to 20.8%, 0.3 p.p. above the path with the design pinned, since a more Bismarckian system dissipates less and is worth more to the electorate at the margin. Under the quadratic cost the design reaches 0.809, and under the linear cost the Bismarckian corner by 2050, after which the design no longer moves.[24]

Figure 7.1: The design and the tax along the baseline path



Note: Log preferences, $\rho = 1$, from the calibration year on; the design chosen at each date applies from the next. The three specifications of the cost are those of table 4, each at its U.S. calibration; the pinned tax path is under the linear-quadratic cost. The values, the paths from the first period and the other elasticities are in the online appendix.

Ageing. Acute ageing, $\nu_t = 1$ throughout, raises the chosen design from 0.738 to 0.874 under the linear-quadratic cost. A scarcer young generation lifts the political weight of retirees, who are unanimous for a more earnings-related system since it increases the labor supply on which their benefits are levied (appendix C.1). And the tax that ageing calls for with the design pinned, 22.3% against 14.4%, raises the loss per unit of redistribution by more than half, since the loss is proportional to τ_t. This channel supplies about half the response under the quadratic cost, where it can be switched off: under a cost on the design itself, which does not scale with the size of the system, the design rises only to 0.778 against 0.825 (online appendix). The chosen design adds 0.4 p.p. to the pinned tax response of 7.9 p.p., which is below the 8.7 p.p. of section 6 because the cost makes a larger system worth less to retirees at the margin.

Income distribution. The model's inequality is the dispersion of relative incomes, not the Gini coefficient of figure 1.1: France's flatter distribution halves $\tilde{V}$, from 0.436 to 0.218, and lowers the amount the flat component carries, M, by a quarter, from 0.575

[24]Recalibrated at Frisch elasticities $\xi = 0.2$ and 0.4 under the quadratic cost, λ moves by about one percent and the tax in 2110 by about 0.3 p.p. either way, while the design in 2110 is unchanged to the third decimal and the acute-ageing design within one unit of it (online appendix).

Table 5: Endogenous pension design across counterfactuals and cost specifications in the U.S.

Scenario	Linear-quadratic			Linear			Quadratic		
	θ chosen	Tax, pinned	Tax, chosen	θ chosen	Tax, pinned	Tax, chosen	θ chosen	Tax, pinned	Tax, chosen
Baseline	0.738	14.43%	14.43%	0.738	14.43%	14.43%	0.738	14.43%	14.43%
Acute ageing	0.874	22.29%	22.73%	1.000	21.97%	23.24%	0.825	22.51%	22.80%
Income distribution	0.850	13.81%	13.91%	1.000	13.85%	14.51%	0.772	13.78%	13.80%
Voting patterns	0.622	15.25%	15.25%	0.047	15.22%	15.01%	0.653	15.27%	15.26%

Note: Log preferences, $\rho = 1$. One block of columns per specification of the cost, each calibrated so that the U.S. electorate re-elects its observed design θ^*; the linear-quadratic cost at $\lambda_1 = 0.674$ and $\lambda_2 = 5.237$. Every row is a separate equilibrium path read in 2020, with the changed characteristic holding over the whole horizon and the political choice binding from the first period. "Pinned" holds the design at θ^* and "chosen" lets the electorate choose it one period in advance, both under the cost. The calibrations are in table 4; the savings rate, the workweek and the other counterfactuals in table 11, the quadratic cost at $\rho = 0.5$ and 2 in table 13, and the UK as host in the online appendix.

to 0.422. With the design pinned it lowers the tax from 14.4% to 13.8%, less than the 1.2 p.p. of section 6, because a lower dispersion also lowers the share of revenue lost, and the tax moves as little under every variant, chosen or pinned. The design is another matter. Under the quadratic cost it moves to 0.772, close to the observed one, since the cost is proportional to the same dispersion as the redistribution a flat benefit performs: compressing the distribution lowers the redistributive stakes and their cost together, and only the young's forward-looking stake, which does not scale with the dispersion, gains a little ground. Under the linear-quadratic cost it moves to 0.850, nearly as far as under acute ageing. The linear term is weighted by the amount carried, which falls by a quarter where $\tilde{V}$ halves, so the cost of redistribution falls by less than the redistributive stake and the forward-looking stake gains more ground; and the linear term has no curvature, so the marginal cost of redistribution falls less as the design becomes more earnings-related than under the quadratic cost, whose curvature damps the response. The tax, 13.9%, is within 0.1 p.p. of the pinned reading.

Voting patterns. France's much flatter voting profile moves the design the other way, to 0.622, by about as much as its income distribution moves it up: when the poor carry more voting weight, the redistributive force of proposition 3 regains ground. The tax rises by 0.8 p.p. in both readings. Combined with France's income distribution, the voting patterns prevail: the two together put the design at 0.641, below the U.S. one, as under the quadratic cost, 0.627 (table 11 in appendix G.2).

As in the cross-section of figure 1.1, ageing moves the chosen design a long way toward earnings-related benefits, and the size of the system with it, under every variant. Inequality moves the size of the system little under every variant, but it moves the design:

under the linear-quadratic cost a compressed income distribution moves it about as far as ageing, toward earnings-related benefits, and France's voting patterns as far the other way. The weak relation between inequality and the design in the cross-section is reproduced by the quadratic cost, under which the design moves little. Under the linear-quadratic cost the design responds to inequality in both directions, and the composite of France's income distribution and voting patterns, the pair the cross-section confronts, leaves it at 0.641, a fall of about 0.10 toward flat benefits in which the voting patterns prevail.

The other two variants show what each ingredient contributes. The Harberger curvature alone, the quadratic cost, damps every response, since its marginal cost falls as the design becomes more earnings-related: acute ageing moves the design to 0.825, France's income distribution to 0.772 and its voting patterns to 0.653, and France's corner is out of reach. Okun's leak alone, the linear cost, prices the first unit of redistribution but has no curvature, so the chosen design responds to a characteristic as a switch: acute ageing and France's income distribution send it to the Bismarckian corner, France's voting patterns to 0.047, next to the Beveridgean one, and the UK's electorate under the U.S. parameter to the Beveridgean corner itself (section 7.2). Together, the leak lets a characteristic move the design further and makes France's corner reachable at a finite parameter, while the curvature keeps the choice interior. The scaling by the size of the system, common to the three, is what carries the ageing response through the tax, shown above for the quadratic cost (online appendix). Mild ageing, the composites, the savings rate and the workweek under each variant are in appendix G.2.

The effect of the intertemporal elasticity of substitution. The linear and the linear-quadratic costs are computed under log preferences only; the quadratic cost is calibrated at each elasticity (appendix G.3). Its parameter falls from 18.267 at $\rho = 0.5$ to 1.724 at $\rho = 2$, as a higher elasticity strengthens the forward-looking channels against the Beveridgean corner; at $\rho = 2$ it is closest to what estimated elasticities imply (appendix C.2), adding to the case of section 5 for $\rho \in [1, 2]$. The ageing response is stable, 0.816 and 0.826 at $\rho = 0.5$ and 2 against 0.825 at $\rho = 1$. The responses to inequality and to voting patterns are not: at $\rho = 2$ the observed design loses only 0.4% of revenue, and on this thin cushion France's income distribution moves the design to 0.918 and its voting patterns to 0.335, against 0.707 and 0.687 at $\rho = 0.5$ (table 13; online appendix).

8 Conclusions

This paper studies how two features of a pension system, whether benefits require a contribution record and how closely they follow past earnings, shape its political support and through it taxes, savings and labor supply. The setting is an overlapping generations model with capital, heterogeneous households, an informal sector and probabilistic voting,

first over the size of the system and then over its design as well.

With log-GHH preferences and hand-to-mouth informal households the equilibrium tax is a function of parameters and demographics alone. It rises with the political weight of the old, with ageing and with income inequality, and it falls when benefits are more closely tied to earnings, since earnings-related benefits move revenue from poor retirees to rich ones; universal pensions raise the tax when informal retirees are poorer than formal ones and the pension is a comparable share of their consumption. Calibrated to Argentina, the model accounts for about three quarters of the rise in pension spending that followed the expansion of coverage of 2005 to 2010, with the savings rate lower by 0.4 p.p. of GDP and the workweek by a quarter of an hour. Calibrated to the U.S., France and the UK, ageing is the main determinant of the tax rate and the earnings link the second; income inequality and voting patterns matter little, and a higher intertemporal elasticity of substitution changes only the size of the earnings-link effect.

When the electorate chooses the earnings link as well, costless redistribution puts the choice at a corner: with propensities to vote that do not rise with income the design chosen with the tax is fully Beveridgean, and at U.S. inequality every timing of the choice gives a corner. An interior design needs flat benefits to be costlier than earnings-related ones. With a deadweight cost of redistribution that rises with the size of the system, calibrated on the U.S. and UK designs, ageing makes the chosen design more Bismarckian and France is predicted more Bismarckian than both; each of the three specifications of the cost we compare at an intertemporal elasticity of one orders the designs of the UK, the U.S. and France as observed. Inequality moves spending little under every specification; it moves the design little when the cost is quadratic in the redistribution performed, and under the linear-quadratic cost a more compressed distribution and a flatter voting profile move it about as far in opposite directions, with the voting profile prevailing when the two are combined, a fall of about 0.10 to 0.641. The pattern of the OECD cross-section, in which spending and the earnings link rise together as population growth falls and neither follows inequality, is that of ageing under every specification and of inequality under the quadratic cost.

Two limitations point to future work. The model has no risk; in Berg and Gonzalez-Eiras (2023) we study a related environment in which health risk may prevent households from qualifying for a contributive pension, and find that households work and save more in response. And one cost of the family we consider does not deliver the three designs at once: a cost that loses a constant share of every unit redistributed lets an electorate choose a fully Bismarckian system, as France's is, but the share France needs puts the UK well below its observed design, so a formality margin, toward which the informal sector of our Argentine model is a first step, remains the structural candidate. The same margin bears on the rise of platform work and other arrangements outside contributive coverage in rich economies, which the model can address.

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Appendices

A Economic Equilibrium

The appendix derives results for the economic equilibrium with CRRA-GHH and log-GHH preferences and where informal workers' endowment is tied to wages in the formal sector absent taxes.

A.1 With CRRA-GHH preferences

Household solutions With CRRA-GHH preferences, the first order conditions in (6) can be written as

$$h_{t,i} = \left(\frac{\eta_i w_t (1-\tau_t) + \frac{\theta \bar{b}_{t+1} \eta_i}{R_{t+1}}}{X_i} \right)^{\xi}$$
$$c^i_{2,t+1} = \tilde{c}^i_{1,t} \left(\beta R_{t+1}\right)^{\rho}.$$

Using the budget constraints and rearranging slightly, we can write individual savings as

$$s_{t,i} = \frac{B_{t+1}}{1+B_{t+1}} \left[w_t \left(1-\tau_t\right) h_{t,i} \eta_i - \frac{X_i \xi}{1+\xi} \left(h_{t,i}\right)^{\frac{1+\xi}{\xi}} \right] - \frac{b^i_{t+1}/R_{t+1}}{1+B_{t+1}},$$

where we have used the shorthand notation $B_{t+1} \equiv (\beta)^{\rho} \left(R_{t+1}\right)^{\rho-1}$.

When informal households save using informal technology, similar conditions apply:[25]

$$h_{t,0} = \left(\frac{\eta_0 w^0_t}{X_0} \right)^{\xi} \tag{A.1}$$
$$c^0_{2,t+1} = \tilde{c}^0_{1,t} \left(\beta R^0_{t+1}\right)^{\rho}. \tag{A.2}$$

As for formal household we can derive the following for informal savings:

$$s_{t,0} = \frac{B^0_{t+1}}{1+B^0_{t+1}} \left[w^0_t h_{t,0} \eta_0 - \frac{X_0 \xi}{1+\xi} \left(h_{t,0}\right)^{\frac{1+\xi}{\xi}} \right] - \frac{b^0_{t+1}/R^0_{t+1}}{1+B^0_{t+1}},$$

where we have used $B^0_{t+1} \equiv (\beta)^{\rho} \left(R^0_{t+1}\right)^{\rho-1}$. When informal households are hand-to-mouth and receive an endowment when old, the solution for labor supply is given by (A.1).

Savings and labor supply in equilibrium. Next, in economic equilibrium, we impose conditions for factor prices in equations (3) and the specification of pension benefits

[25] Note that given our assumption on universal pensions, informal workers do not perceive a link between their work effort and the benefits they might receive.

including the balanced budget condition in equations (8). We can then rewrite type i's labor supply decision as

$$h_{t,i} = \left(\frac{\eta_i}{X_i}\right)^{\xi} \left(\frac{1}{h_t}\right)^{\xi} \left((1-\alpha)(1-\tau_t)\left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha} h_t^{1-\alpha} + \frac{1-\alpha}{\alpha}\frac{\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1}\right)^{\xi}.$$

The only type-specific term is $(\eta_i/X_i)^{\xi}$; this shows that the ratio $h_{t,i}/h_t$ is only a function of parameters, namely that

$$\frac{h_{t,i}}{h_t} = \frac{(\eta_i/X_i)^{\xi}}{\Gamma_h}, \qquad \Gamma_h \equiv \sum_{i>0} \frac{\gamma_i \eta_i^{1+\xi}}{X_i^{\xi}}. \tag{A.3}$$

Next, we can use the individual labor decision and wage rates to write

$$(1-\alpha)(1-\tau_t) h_t^{1-\alpha} \left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha} = \frac{h_t^{\frac{1+\xi}{\xi}}}{\Gamma_h^{1/\xi}} - \frac{1-\alpha}{\alpha}\frac{\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1},$$

Using this and the discounted pension benefits in equilibrium, we have that

$$\begin{aligned} s_{t,i} =& \frac{1}{1+B_{t+1}} \left[B_{t+1} \frac{\eta_i^{1+\xi}}{X_i^{\xi}} \left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \frac{1}{1+\xi} - \frac{1-\alpha}{\alpha}\frac{1-\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1} \right] \\ & - \frac{\eta_i^{1+\xi}}{X_i^{\xi}\Gamma_h} \frac{1-\alpha}{\alpha}\frac{\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1} \end{aligned} \tag{A.4}$$

Using the definition of the aggregate state (1) and rearranging, we then have:

$$s_t = \left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \Gamma_{s,t}, \qquad \Gamma_{s,t} \equiv \frac{1}{1+\xi} \frac{\Gamma_h B_{t+1}}{1 + B_{t+1} + \frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta B_{t+1})} \tag{A.5}$$

Furthermore, note that we can combine equations for s_t, h_t to yield a closed form expressions *given* values of B_{t+1}. From the definition of h_t we use that s_t is proportional to $h_t^{(1+\xi)/\xi}$ to derive:

$$h_t = \Theta_{h,t} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha\xi}{1+\alpha\xi}}, \qquad \Theta_{h,t} \equiv (\Gamma_h)^{\frac{1+\xi}{1+\alpha\xi}} \left(\frac{(1-\alpha)(1-\tau_t)}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}\right)^{\frac{\xi}{1+\alpha\xi}} \tag{A.6}$$

Using this, an alternative expression for s_t is:

$$s_t = \Theta_{s,t} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}}, \qquad \Theta_{s,t} \equiv \left(\frac{\Theta_{h,t}}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \Gamma_{s,t}. \tag{A.7}$$

Using the relation between s_t and $h_t^{\frac{1+\xi}{\xi}}$ in (A.6), the ratio of individual (A.4) to aggregate

savings satisfies:

$$\frac{s_{t,i}}{s_t} = \frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{1}{\Gamma_h} + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}}{1+\gamma_0\epsilon} \frac{1-\theta}{1+B_{t+1}} \left(\frac{\eta_i^{1+\xi}}{X_i^{\xi}\Gamma_h} - 1 \right). \tag{A.8}$$

For informal households, we assume that informal wages are proportional to formal wages absent taxes and scaled appropriately by informal productivity η_0. Furthermore, we assume that the return to informal savings is proportional to the formal return on savings $R_t^0 = r_0 R_t$.

Specifically, we assume that informal wages are given by:

$$w_t^0 = \left(\frac{1-\alpha}{\Gamma_h^{\alpha}} \right)^{\frac{1}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t} \right)^{\frac{\alpha}{1+\alpha\xi}}$$

Informal labor supply then follows directly from the derivations above for formal workers; in the case with informal savings, the ratio of informal to aggregate savings is similarly given by

$$\frac{s_{t,0}}{s_t} = \frac{B_{t+1}^0}{1+B_{t+1}^0} \frac{\eta_0^{1+\xi}}{X_0^{\xi}(1+\xi)} \left(\frac{1-\alpha}{\Gamma_h^{\alpha}} \right)^{\frac{1+\xi}{1+\alpha\xi}} \frac{1}{\Theta_{s,t}} - \frac{1}{1+B_{t+1}^0} \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_{t+1}}{1+\gamma_0\epsilon} \tag{A.9}$$

Consumption functions. It is straightforward to derive relevant equilibrium consumption functions $(c_{1,t}^j, \tilde{c}_{1,t}^j, c_{2,t}^j)$ given the results above. Imposing household budget constraints (5), factor prices (3), pension benefits and balanced budget condition (8), and labor and savings decisions from above, we have for formal households i:

$$c_{1,t}^i = \frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{s_t}{\Gamma_{s,t}} \left(1 - \frac{B_{t+1}}{(1+B_{t+1})(1+\xi)} \right) + \frac{s_t}{1+B_{t+1}} \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon} \tag{A.10}$$

$$c_{2,t}^i = \alpha\nu_t h_t^{1-\alpha} \left(\frac{s_{t-1}}{\nu_t} \right)^{\alpha} \left(\frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha} \frac{\tau_t}{1+\gamma_0\epsilon} \left(1 + \theta \left[\frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{1}{\Gamma_h} - 1 \right] \right) \right) \tag{A.11}$$

$$\tilde{c}_{1,t}^i = \frac{s_t}{1+B_{t+1}} \left(\frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{1}{(1+\xi)\Gamma_{s,t}} + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon} \right) \tag{A.12}$$

When informal households save using informal technology, equilibrium consumption functions are given by

$$\tilde{c}_{1,t}^0 = \frac{1}{1+\xi} \frac{\eta_0^{1+\xi}}{X_0^{\xi}} \left(\frac{1-\alpha}{\Gamma_h^{\alpha}} \right)^{\frac{1+\xi}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t} \right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} + \frac{1}{1+B_{t+1}^0} \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_{t+1}}{1+\gamma_0\epsilon} s_t \tag{A.13}$$

$$c_{2,t}^0 = r_0\alpha\nu_t \left(\frac{s_{t-1}}{\nu_t} \right)^{\alpha} h_t^{1-\alpha} \left(\frac{s_{t-1,0}}{s_{t-1}} + \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_t}{1+\gamma_0\epsilon} \right). \tag{A.14}$$

When informal households are hand-to-mouth, relevant equilibrium consumption func-

tions are given by[26]

$$\tilde{c}^0_{1,t} = \frac{1}{1+\xi}\frac{\eta_0^{1+\xi}}{X_0^{\xi}}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}\left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} \tag{A.15}$$

$$\tilde{c}^0_{2,t} = \frac{1}{1+\xi}\frac{(\chi\eta_0)^{1+\xi}}{X_0^{\xi}}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}\left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} + \frac{(1-\alpha)\nu_t\epsilon\tau_t}{1+\gamma_0\epsilon}\left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha} h_t^{1-\alpha}. \tag{A.16}$$

A.2 With log-GHH preferences

With log-preferences, the economic equilibrium is identical to the CRRA equations outlined in appendix A.1, except for the discount function $B_t = B_t^0 = \beta$. All relevant variables $x_t \in \{s_t, h_t, s_t^i, h_t^j, \tilde{c}_{1,t}^j, c_{2,t}^i, \tilde{c}_{2,t}^0, c_{2,t+1}^i, \tilde{c}_{2,t+1}^0\}$ can be written on the form $x_t = \Theta_{x,t}\left(\frac{s_{t-1}}{\nu_t}\right)^{\sigma_x}$ where σ_x are parameter functions

$$\begin{aligned}
\sigma_h = \sigma_h^j &= \frac{\alpha\xi}{1+\alpha\xi} \\
\sigma^j_{\tilde{c}_1,t} = \sigma^i_{c_2,t} = \sigma^0_{\tilde{c}_2,t} = \sigma_{s,t} &= \sigma^i_{s,t} = \frac{\alpha(1+\xi)}{1+\alpha\xi} \\
\sigma^i_{c_2,t+1} = \sigma^0_{\tilde{c}_2,t+1} &= \left(\frac{\alpha(1+\xi)}{1+\alpha\xi}\right)^2,
\end{aligned}$$

and where functions $\Theta_{x,t}$ only depends on parameters and policy.

Proof of proposition 1. Part *i.* outlines the marginal effect on s_t, h_t from policy and pension characteristics. Given the solution structure, the effect on s_t, h_t can be found by differentiating equations (A.5) and (A.6) with respect to the relevant variable. For τ_t the effect is straightforward:

$$\begin{aligned}
\frac{\partial \ln(h_t)}{\partial \tau_t} &= -\frac{\xi}{1+\alpha\xi}\frac{1}{1-\tau_t} \leq 0 \\
\frac{\partial \ln(s_t)}{\partial \tau_t} &= -\frac{1+\xi}{1+\alpha\xi}\frac{1}{1-\tau_t} \leq 0.
\end{aligned}$$

We can also show that h_t is unambiguously decreasing in ϵ:

$$\begin{aligned}
\frac{\partial \ln(h_t)}{\partial \epsilon} &= \frac{\xi}{1+\alpha\xi}\frac{\frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}\frac{1}{1+\gamma_0\epsilon}\left((1+\gamma_0\epsilon)\frac{\partial \ln(\Gamma_{s,t})}{\partial \epsilon} - \gamma_0\right) \leq 0 \\
\frac{\partial \ln(\Gamma_{s,t})}{\partial \epsilon} &= \frac{\gamma_0}{1+\gamma_0\epsilon}\frac{\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta)}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta)}.
\end{aligned}$$

[26] Recall informal workers are assumed to receive an endowment proportional to wages of formal workers when they retire. For tractability we define this as if they were making a labor supply decision. The relevant parameter determining the size of the endowment is χ.

However, the effect on s_t is harder to sign.

$$\frac{\partial \ln(s_t)}{\partial \epsilon} = \frac{1+\xi}{\xi}\frac{\partial \ln(h_t)}{\partial \epsilon} + \frac{\partial \ln(\Gamma_{s,t})}{\partial \epsilon}.$$

Replacing from the expressions above, and neglecting common factors, the sign of the derivative is given by the sign of

$$-\frac{1}{1+\alpha\xi}\frac{\theta\beta(1+\beta)}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta\frac{\xi}{1+\xi})} + 1 + \theta\beta.$$

This can be seen to be positive for all $\theta \in [0,1]$. Note that the effect from γ_0 is equivalent to that of ϵ. We now consider the effect of future taxes on labor supply and savings.

$$\frac{\partial \ln(h_t)}{\partial \tau_{t+1}} = \frac{\xi}{1+\alpha\xi}\frac{\frac{1-\alpha}{\alpha}\frac{\theta\Gamma_{s,t}}{1+\gamma_0\epsilon}}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}\left(1+\tau_{t+1}\frac{\partial \ln(\Gamma_{s,t})}{\partial \tau_{t+1}}\right) \geq 0$$

$$\tau_{t+1}\frac{\partial \ln(\Gamma_{s,t})}{\partial \tau_{t+1}} = -\frac{\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1+\theta\beta)}{1+\gamma_0\epsilon}}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1+\theta\beta)}{1+\gamma_0\epsilon}} \in (-1,0].$$

This shows that labor supply is increasing in τ_{t+1}. For savings the effect is:

$$\frac{\partial \ln(s_t)}{\partial \tau_{t+1}} = \frac{1+\xi}{\xi}\frac{\partial \ln(h_t)}{\partial \tau_{t+1}} + \frac{\partial \ln(\Gamma_{s,t})}{\partial \tau_{t+1}}.$$

As before the sign of this derivative can be shown to be the same as the sign of the following term:

$$\frac{1}{1+\alpha\xi}\frac{\theta\beta(1+\beta)}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta\frac{\xi}{1+\xi})} - (1+\theta\beta).$$

This is basically the same term (with different sign) as before. Thus, for all values of θ and τ_{t+1} we have that an increase in τ_{t+1} reduces savings.

Taking the derivative with respect to θ, we see a similar condition as with the effect of τ_{t+1}, namely:

$$\frac{\partial \ln(h_t)}{\partial \theta} = \frac{\xi}{1+\alpha\xi}\frac{\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}\Gamma_{s,t}}{1+\gamma_0\epsilon}}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}\left(1+\theta\frac{\partial \ln(\Gamma_{s,t})}{\partial \theta}\right) \geq 0,$$

$$\theta\frac{\partial \ln(\Gamma_{s,t})}{\partial \theta} = -\frac{\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}\theta\beta}{1+\gamma_0\epsilon}}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1+\theta\beta)}{1+\gamma_0\epsilon}} \in (-1,0].$$

This shows that labor supply is increasing in θ. For the savings rate, we have:

$$\frac{\partial \ln(s_t)}{\partial \theta} = \frac{1+\xi}{\xi}\frac{\partial \ln(h_t)}{\partial \theta} + \frac{\partial \ln(\Gamma_{s,t})}{\partial \theta}.$$

Proceeding as before we find that this is unambiguously negative.[27]

Part *ii.* follows from earlier derivations, specifically equation (A.3). Part *iii.* similarly follows from equation (A.8) with $B_{t+1} = \beta$ in the log case. Finally, part *iv.* follows straightforwardly, comparing (A.8) and (A.3): An average type i household has $\eta_i^{1+\xi}/X_i^{\xi} = \Gamma_h$. Thus, for $\tau_{t+1} = 0$ or $\theta = 1$, we simply have $s_{t,i}/s_t = \eta_i h_{t,i}/h_t$. The spread in savings rates increase in τ_{t+1} and decrease in θ in the sense that these parameter changes increase $s_{t,i}/s_t$ for $s_{t,i}/s_t > 1$ and decrease for $s_{t,i}/s_t < 1$.

Part *v.* The effects on $s_{t,0}$ and $s_{t,0}/s_t$ from various parameters come from differentiation of (A.9) with $B^0_{t+1} = \beta$. Note that $s_{t,0}$ only depends on τ_t though the effect of taxes on s_t. Thus, it is straightforward to show that both $s_{t,0}$ and $s_{t,0}/s_t$ are increasing in τ_t.

The effects from τ_{t+1} and ϵ are given by:

$$\frac{\partial s_{t,0}}{\partial \tau_{t+1}} = -\frac{1}{1+\beta}\frac{1-\alpha}{r_0\alpha}\frac{s_t\epsilon}{1+\gamma_0\epsilon}\left(1+\tau_{t+1}\frac{\partial \ln(s_t)}{\partial \tau_{t+1}}\right) \leq 0,$$
$$\frac{\partial s_{t,0}}{\partial \epsilon} = -\frac{1}{1+\beta}\frac{1-\alpha}{r_0\alpha}\frac{s_t\tau_{t+1}}{1+\gamma_0\epsilon}\left(\frac{1}{1+\gamma_0\epsilon}+\epsilon\frac{\partial \ln(s_t)}{\partial \epsilon}\right) \leq 0.$$

Where the first derivative can be signed since $\tau_{t+1}\frac{\partial \ln(s_t)}{\partial \tau_{t+1}} \geq -1$. It is straightforward to show that an increase in θ increases $s_{t,0}$.

To see the effect of τ_{t+1} on the ratio $s_{t,0}/s_t$ we have that:

$$\frac{\partial(s_{t,0}/s_t)}{\partial \tau_{t+1}} = -\frac{\beta}{1+\beta}\frac{\eta_0^{1+\xi}}{X_0^{\xi}(1+\xi)}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}\frac{1}{\Theta_{s,t}}\frac{\partial \ln(\Theta_{s,t})}{\partial \tau_{t+1}} - \frac{1}{1+\beta}\frac{1-\alpha}{r_0\alpha}\frac{\epsilon}{1+\gamma_0\epsilon},$$

where the derivative $\partial \ln(\Theta_{s,t})/\partial \tau_{t+1} = \partial \ln(s_t)/\partial \tau_{t+1}$ is defined above. We note that this may be negative or positive. For instance, if $\epsilon \approx 0$ an increase in future taxes increases this ratio. Conversely if $r_0 \approx 0$ then we would expect the opposite effect.

B Politico-economic equilibrium

B.1 Proof of proposition 2

As shown before, in equilibrium, all consumption functions are log-separable in the aggregate state s_{t-1}, h_{t-1} (see appendix A.1). In all the Θ-coefficient functions, the only *predetermined* variables that may enter the political problem as an endogenous state, is the distribution of savings $s_{t-1,i}/s_{t-1}$. However, the current distribution, $s_{t,i}/s_t$, is

[27] The term whose sign determines this is slightly different

$$\frac{1}{1+\alpha\xi}\frac{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta\frac{\xi}{1+\xi})} - 1.$$

solely a function of parameters and the *current* policy. Thus, while $\{s_{t-1,i}/s_{t-1}\}_{i>0} \in z_t$, $\{s_{t,i}/s_t\}_{i>0}$ is independent of τ_t implying that

$$\frac{\partial z_{t+1}}{\partial \tau_t} = \frac{\partial \zeta_t}{\partial \tau_t} = 0,$$

where ζ_t is the law of motion for states in the political objective (eq. 9). Thus, we say that this distribution is not an *endogenous* state, as there are no strategic effects from changing τ_t on the future policy τ_{t+1}.

As there are no endogenous states, the politico-economic equilibrium is characterized by the first order condition

$$\omega\left[\gamma_0 \overbrace{\frac{\partial \ln\left(\Theta^0_{\tilde{c}_2,t}\right)}{\partial \tau_t}}^{=\mathcal{E}^0_{2,t}} + \sum_i \gamma_i \overbrace{\frac{\partial \ln\left(\Theta^i_{c_2,t}\right)}{\partial \tau_t}}^{=\mathcal{E}^i_{2,t}}\right]$$
$$+\nu_t\left\{\gamma_0 \underbrace{\left(\frac{\partial \ln\left(\Theta^0_{\tilde{c}_1,t}\right)}{\partial \tau_t} + \beta \frac{\partial \ln\left(\Theta^0_{\tilde{c}_2,t+1}\right)}{\partial \tau_t}\right)}_{=\mathcal{E}^0_{1,t}} + \sum_i \gamma_i \underbrace{\left(\frac{\partial \ln\left(\Theta^i_{\tilde{c}_1,t}\right)}{\partial \tau_t} + \beta \frac{\partial \ln\left(\Theta^i_{c_2,t+1}\right)}{\partial \tau_t}\right)}_{=\mathcal{E}^i_{1,t}}\right\} \leq 0,$$

with equality for an interior solution. We introduce the shorthand notation $\mathcal{E}^j_{x,t}$ to denote the marginal effect of taxes on the indirect utility of generation x of type j.

Using equilibrium consumption functions from appendix A.1, we have

$$\mathcal{E}^0_{2,t} = \frac{\frac{(1-\alpha)\nu_t\epsilon}{1+\gamma_0\epsilon}\Theta_{h,t}^{1-\alpha}\left(1+(1-\alpha)\tau_t\frac{\partial \ln(\Theta_{h,t})}{\partial \tau_t}\right)}{\frac{(\chi\eta_0)^{1+\xi}}{(1+\xi)X_0^\xi}\left(\frac{1-\alpha}{\Gamma_h^\alpha}\right)^{\frac{1+\xi}{1+\alpha\xi}} + \frac{(1-\alpha)\nu_t\epsilon\tau_t}{1+\gamma_0\epsilon}\Theta_{h,t}^{1-\alpha}} \tag{B.1}$$

$$\mathcal{E}^i_{2,t} = (1-\alpha)\frac{\partial \ln(\Theta_{h,t})}{\partial \tau_t} + \frac{\frac{1-\alpha}{\alpha}\frac{1}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_i^{1+\xi}}{X_i^\xi}\frac{1}{\Gamma_h}-1\right]\right)}{\frac{s^i_{t-1}}{s_{t-1}} + \frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_i^{1+\xi}}{X_i^\xi}\frac{1}{\Gamma_h}-1\right]\right)} \tag{B.2}$$

$$\mathcal{E}^0_{1,t} = -\frac{\beta}{1-\tau_t}\frac{\alpha(1+\xi)^2}{(1+\alpha\xi)^2} \tag{B.3}$$

$$\mathcal{E}^i_{1,t} = -\frac{1}{1-\tau_t}\frac{1+\xi}{1+\alpha\xi}(1+\beta), \tag{B.4}$$

It is straightforward to verify that all young households' indirect utility is decreasing in τ_t. For all retirees, there are opposing effects from adjusting the tax rate: The first term is negative and works through the labor response from current working households ($w_t h_t$ decreases with τ_t) on the pension contributions. The second term is positive and relates to the direct effect of higher pension benefits and the increase in the interest rate.

In an interior equilibrium it must be the case that retirees prefer positive taxes, as

the young unambiguously always support zero taxes. Then, increasing ω or decreasing ν_t naturally leads to higher equilibrium tax rates. In the corner with $\tau_t = 0$, there will be no effect from marginal changes in ω, ν_t. Thus $\partial\tau_t/\partial\omega \geq 0$ and $\partial\tau_t/\partial\nu_t \leq 0$. The sign in ν_t needs one more step, since ν_t also scales the universal pension relative to the endowment of informal retirees and so raises their support: with $\sigma_t^0 \equiv b_t^0/\tilde{c}_{2,t}^0$ the pension's share of their consumption, $\partial\mathcal{E}_{2,t}^0/\partial\nu_t = \mathcal{E}_{2,t}^0(1-\sigma_t^0)/\nu_t$. Writing z_t for the left-hand side of the first order condition and using the condition itself to substitute for the young's terms, $\partial z_t/\partial\nu_t = -\frac{\omega}{\nu_t}\left[\gamma_0\mathcal{E}_{2,t}^0\,\sigma_t^0 + \sum_{i>0}\gamma_i\mathcal{E}_{2,t}^i\right]$, which is negative whenever the aggregate marginal support of formal retirees, $\sum_{i>0}\gamma_i\mathcal{E}_{2,t}^i$, is positive at the optimum, that is, unless the tax is sustained by informal retirees against formal ones.

To analyze the effect of income inequality on equilibrium taxes, without loss of generality we consider two types, H and L, with productivities above and below the average respectively. We increase the productivity for type H and that of type L such that

$$\frac{\partial\eta_L}{\partial\eta_H} = -\frac{\gamma_H\left(\frac{\eta_H}{X_H}\right)^\xi}{\gamma_L\left(\frac{\eta_L}{X_L}\right)^\xi},$$

which keeps the average labor supply and savings (and Γ_h) constant for a given path of policies. We note that for young generation, this does not alter the determinants of taxes as $\mathcal{E}_{1,t}^i$ is unaffected by this. The effect on relevant retirees $m = L, H$ are given by:

$$\frac{\partial\mathcal{E}_{2,t}^m}{\partial\eta_m} = -(1+\xi)\left(\frac{\eta_m}{X_m}\right)^\xi\frac{1}{\Gamma_h}\frac{1-\alpha}{\alpha}\frac{1}{1+\gamma_0\epsilon}\frac{(1-\theta)\left(1+\frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\frac{1}{1+\beta}\right)}{\left(\frac{s_{t-1}^m}{s_{t-1}}+\frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_m^{1+\xi}}{X_m^\xi}\frac{1}{\Gamma_h}-1\right]\right)\right)^2}$$

This shows that relatively high income households have a weaker preference for taxes. Using this, it is now straightforward to show that $\partial(\gamma_L\mathcal{E}_{2,t}^L + \gamma_H\mathcal{E}_{2,t}^H)/\partial\eta_H > 0$. This is driven by the fact that the marginal utility of consumption is higher for low income households. Thus, higher inequality increases equilibrium taxes.

A similar argument can be applied to see the marginal effects on types H and L of an increase in θ:

$$\frac{\partial\mathcal{E}_{2,t}^m}{\partial\theta} = \frac{\frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left[\frac{\eta_m^{1+\xi}}{X_m^\xi}\frac{1}{\Gamma_h}-1\right]}{\frac{s_{t-1}^m}{s_{t-1}}+\frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_m^{1+\xi}}{X_m^\xi}\frac{1}{\Gamma_h}-1\right]\right)}\frac{s_{t-1}^m/s_{t-1}}{\frac{s_{t-1}^m}{s_{t-1}}+\frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_m^{1+\xi}}{X_m^\xi}\frac{1}{\Gamma_h}-1\right]\right)}$$

Thus, an increase in θ increases the support for taxes among the high income households but weakens it among the poor. For the same reasons as above, namely the higher marginal utility of poor households, higher θ then lowers equilibrium tax rates.

The effect of ϵ works through retirees alone, since $\mathcal{E}_{1,t}^0$ and $\mathcal{E}_{1,t}^i$ do not depend on ϵ. Write $\sigma_t^j \equiv b_t^j/c_{2,t}^j$ for the pension's share of the consumption of a retiree of type j, as σ_t^0

above, and $\phi_t \equiv 1 + (1-\alpha)\tau_t \, \partial \ln(\Theta_{h,t})/\partial \tau_t$ for the marginal revenue of the tax relative to its base. The expressions above are then $\mathcal{E}^0_{2,t} = \phi_t \sigma^0_t / \tau_t$ and $\mathcal{E}^i_{2,t} = \Big(\phi_t \sigma^i_t - (1-\phi_t)(1-\sigma^i_t)\Big)/\tau_t$: a unit of tax raises every pension by ϕ_t/τ_t in proportion, and with log utility a retiree's support is that proportion times the pension's share of consumption, less, for formal retirees, the fall in the interest rate on their savings. Benefits are shares of revenue $\nu_t w_t h_t \tau_t$, $\epsilon/(1+\gamma_0\epsilon)$ per informal retiree and $(1+\theta_t[y_i - 1])/(1+\gamma_0\epsilon)$ per formal retiree of type i with $y_i \equiv \frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{1}{\Gamma_h}$, so a higher ϵ raises b^0_t by $1/(\epsilon(1+\gamma_0\epsilon))$ in proportion and lowers every b^i_t by $\gamma_0/(1+\gamma_0\epsilon)$ in proportion, a pure transfer since there are γ_0 informal retirees per formal one. With $s_{t-1,i}/s_{t-1}$ and $\Theta_{h,t}$ held fixed as in the comparative statics above, ϵ enters the first order condition only through the shares, and a share responds to a proportional change in the benefit by $\partial \sigma^j_t / \partial \ln b^j_t = \sigma^j_t(1-\sigma^j_t)$, since the rest of consumption does not move. Hence

$$\frac{\partial z_t}{\partial \epsilon} = \frac{\omega \, \gamma_0}{\tau_t(1+\gamma_0\epsilon)} \left[\frac{\phi_t}{\epsilon} \, \sigma^0_t\Big(1-\sigma^0_t\Big) - \sum_{i>0} \gamma_i \, \sigma^i_t\Big(1-\sigma^i_t\Big) \right].$$

At an interior equilibrium z_t is decreasing in τ_t, so taxes rise with ϵ if and only if the bracket is positive,

$$\frac{\phi_t}{\epsilon} \, \sigma^0_t\Big(1-\sigma^0_t\Big) > \sum_{i>0} \gamma_i \, \sigma^i_t\Big(1-\sigma^i_t\Big). \tag{B.5}$$

The share γ_0 multiplies the whole expression, because the loss of formal retirees is the gain of informal retirees times their number, and enters the bracket only through the consumption levels, so it does not determine the sign. The factor $\sigma(1-\sigma)$ is zero for a retiree without a pension and for one who lives on the pension alone, whose consumption and marginal benefit a higher ϵ raises in the same proportion, and is largest at one half. The factor $1/\epsilon$ is the asymmetry of the transfer: a unit of ϵ is a large proportional change of a small universal pension and a small one of the formal pensions. The condition therefore holds when informal retirees are poorer than formal ones and the pension is a comparable share of their consumption, since a comparable share of a smaller consumption is a smaller pension, and fails when they live on the pension alone or are much richer than formal retirees.

B.2 With log-GHH preferences and informal savings

The introduction of $s_{t-1,0}/s_{t-1}$ as an endogenous state variable complicates the model because this state variable not only depends on past choices (through the effect of τ_{t-1} on $\Theta_{s,t-1}$), but is also forward looking ($s_{t-1,0}/s_{t-1}$ depends explicitly on τ_t). To deal with this, let $dx_t/d\tau_t$ denote the effect in politico-economic equilibrium of changing the tax rate τ_t on the variable x_t accounting for the fact that a change in τ_t also changes

the state $s_{t,0}/s_t$ which through the continuation policy affects the choice of τ_{t+1} as well. Specifically, for the state variable $s_{t,0}/s_t$ we have:

$$\frac{d(s_{t,0}/s_t)}{d\tau_t} = \frac{\partial(s_{t,0}/s_t)}{\partial\tau_t} + \frac{\partial(s_{t,0}/s_t)}{\partial\tau_{t+1}}\frac{d\tau^{t+1}}{d(s_{t,0}/s_t)}\frac{d(s_{t,0}/s_t)}{d\tau_t},$$

where the partial derivatives of $s_{t,0}/s_t$ captures how the state reacts to policy in economic equilibrium and $\tau^{t+1}(s_{t,0}/s_t)$ is the continuation policy that details how policy in $t+1$ adjusts when the endogenous state changes. When considering a marginal change in τ_t, the policymaker takes into account how this propagates via economic equilibrium and continuation policy functions into future periods in order to summarize the total impact on current relevant variables. The technical documentation derives a recursive structure for all the relevant derivatives required to evaluate the marginal effect of τ_t on the indirect utility of households (Berg and Gonzalez-Eiras 2026, section *Informal Savings Model*).

With the endogenous state $s_{t-1,0}/s_{t-1}$, the politico-economic equilibrium is now characterized by the first order condition:

$$\omega\left[\sum_{i\geq 0}\gamma_i\overbrace{\frac{d\ln\left(c^i_{2,t}\right)}{d\tau_t}}^{=\mathcal{E}^i_{2,t}}\right] + \nu_t\left\{\sum_{i\geq 0}\gamma_i\underbrace{\left(\frac{d\ln\left(\tilde{c}^i_{1,t}\right)}{d\tau_t} + \beta\frac{d\ln\left(c^i_{2,t+1}\right)}{d\tau_t}\right)}_{=\mathcal{E}^i_{1,t}}\right\} \leq 0,$$

Using equilibrium consumption functions from appendix A.1, we have the derivatives of indirect utility for retired households:

$$\mathcal{E}^0_{2,t} = (1-\alpha)\frac{d\ln(h_t)}{d\tau_t} + \frac{\frac{1-\alpha}{r_0\alpha}\frac{\epsilon}{1+\gamma_0\epsilon}}{\frac{s_{t-1,0}}{s_{t-1}} + \frac{1-\alpha}{r_0\alpha}\frac{\epsilon\tau_t}{1+\gamma_0\epsilon}} \tag{B.6}$$

$$\mathcal{E}^i_{2,t} = (1-\alpha)\frac{d\ln(h_t)}{d\tau_t} + \frac{\frac{1-\alpha}{\alpha}\frac{1}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_i^{1+\xi}}{X_i^{\xi}}\frac{1}{\Gamma_h}-1\right]\right)}{\frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left(1+\theta\left[\frac{\eta_i^{1+\xi}}{X_i^{\xi}}\frac{1}{\Gamma_h}-1\right]\right)}. \tag{B.7}$$

We note that the marginal effect for formal retirees is the same as when continuation policies were independent of state variables. The only difference is that the change in the aggregate labor supply, $\frac{d\ln(h_t)}{d\tau_t}$, now includes strategic effects through future policies.

For the formal, working households, we can use the Euler equation to write the indirect utility as $\ln\left(\tilde{c}^i_{1,t}\right)(1+\beta) + \beta\ln(\beta R_{t+1})$. The marginal effect on indirect utility then consists of the two terms:

$$\mathcal{E}^i_{1,t} = (1+\beta)\frac{d\ln\left(\tilde{c}^i_{1,t}\right)}{d\tau_t} + \beta\frac{d\ln(R_{t+1})}{d\tau_t}$$

$$\frac{d\ln\left(\tilde{c}^i_{1,t}\right)}{d\tau_t} = \frac{1+\xi}{\xi}\frac{d\ln\left(h_t\right)}{d\tau_t} + \frac{\Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{1-\theta_{t+1}}{1+\gamma_0\epsilon}\left(\frac{d\tau_{t+1}}{d\tau_t} + \tau_{t+1}\frac{d\ln(\Gamma_{s,t})}{d\tau_t}\right)}{\frac{\eta_i^{1+\xi}}{X_i^{\xi}}\frac{1}{1+\xi} + \Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon}}$$

$$\frac{d\ln\left(R_{t+1}\right)}{d\tau_t} = (1-\alpha)\left(\frac{d\ln\left(h_{t+1}\right)}{d\ln\left(s_t\right)} - 1\right)\frac{d\ln\left(s_t\right)}{d\tau_t}.$$

where we use the shorthand $d\tau_{t+1}/d\tau_t = \frac{d\tau^{t+1}}{d(s^0_t/s_t)}\frac{d(s^0_t/s_t)}{d\tau_t}$ to define the total effect of a change in the tax rate on future taxes. The technical documentation gives the computation of these derivatives in the numerical application (Berg and Gonzalez-Eiras 2026, section *Informal Savings Model*).

For the informal, working households, a similar Euler condition applies such that we can write indirect utility as $\ln\left(\tilde{c}^0_{1,t}\right)(1+\beta)+\beta\ln\left(\beta R^0_{t+1}\right)$ The marginal effect on indirect utility is then characterized by

$$\mathcal{E}^0_{1,t} = (1+\beta)\frac{d\ln\left(\tilde{c}^0_{1,t}\right)}{d\tau_t} + \beta\frac{d\ln\left(R^0_{t+1}\right)}{d\tau_t}$$

$$\frac{d\ln\left(\tilde{c}^0_{1,t}\right)}{d\tau_t} = \frac{\frac{1-\alpha}{r_0\alpha}\frac{\epsilon}{1+\gamma_0\epsilon}\Theta_{s,t}\left(\frac{d\tau_{t+1}}{d\tau_t} + \tau_{t+1}\frac{d\ln(s_t)}{d\tau_t}\right)}{\frac{\eta_0^{1+\xi}}{X_0^{\xi}}\frac{1}{1+\xi}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}} + \frac{1-\alpha}{r_0\alpha}\frac{\epsilon\tau_{t+1}}{1+\gamma_0\epsilon}\Theta_{s,t}}$$

$$\frac{d\ln\left(R^0_{t+1}\right)}{d\tau_t} = (1-\alpha)\left(\frac{d\ln\left(h_{t+1}\right)}{d\ln\left(s_t\right)} - 1\right)\frac{d\ln\left(s_t\right)}{d\tau_t}$$

B.3 With CRRA-GHH preferences and informal savings

With CRRA preferences it is no longer the case that future policy is independent of the capital stock. But conditional on this state variable, with GHH preferences it is still the case that the distribution of savings for formal workers, $\{s_{t-1,i}/s_{t-1}\}_{i>0}$, remains a function of s_{t-1} and τ_t and is thus not an independent state variable. Thus, the policy function can be written as a function of two endogenous states: s_{t-1} and $s_{t-1,0}/s_{t-1}$.

Compared to the log-GHH case, we adjust the definition of the politico-economic equilibria derivative $dx_t/d\tau_t$ to include the effect through both states such that

$$\frac{dx_t}{d\tau_t} = \frac{\partial x_t}{\partial \tau_t} + \frac{\partial x_t}{\partial \tau_{t+1}}\left(\frac{\partial \tau^{t+1}}{\partial s_t}\frac{ds_t}{d\tau_t} + \frac{\partial \tau^{t+1}}{\partial(s_{t,0}/s_t)}\frac{d(s_{t,0}/s_t)}{d\tau_t}\right).$$

The first order condition identifying the politico-economic choice τ_t is now now given by

$$\omega\sum_{j\geq 0}\gamma_j\underbrace{\left(c^j_{2,t}\right)^{\frac{\rho-1}{\rho}}\frac{d\ln\left(c^j_{2,t}\right)}{d\tau_t}}_{\equiv\mathcal{E}^j_{2,t}} + \nu_t\sum_{j\geq 0}\gamma_j\underbrace{\left(\left(\tilde{c}^j_{1,t}\right)^{\frac{\rho-1}{\rho}}\frac{d\ln\left(\tilde{c}^j_{1,t}\right)}{d\tau_t} + \beta\left(c^j_{2,t+1}\right)^{\frac{\rho-1}{\rho}}\frac{d\ln\left(c^j_{2,t+1}\right)}{d\tau_t}\right)}_{\equiv\mathcal{E}^j_{1,t}} \leq 0,$$

The log-derivatives $d\ln\left(c_{2,t}^{j}\right)/d\tau_t$ follow from the formulas for the log-GHH case and consumption levels $c_{2,t}^{j}$ depend on the state s_{t-1} and are given in appendix A.1.

For formal working households, we use the Euler equation to write indirect utility as

$$\begin{aligned}
v_{1,t}^{i} =&(1+B_{t+1})\frac{\left(\tilde{c}_{1,t}^{i}\right)^{1-1/\rho}}{1-1/\rho}=\frac{\left(\hat{c}_{1,t}^{i}\right)^{1-1/\rho}}{1-1/\rho}\\
\hat{c}_{1,t}^{i} =&\left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}}(1+B_{t+1})^{\frac{1}{\rho-1}}\mathcal{C}_{1,t}^{i}\\
\mathcal{C}_{1,t}^{i} =&\frac{\eta_i^{1+\xi}}{X_i^{\xi}}\frac{1}{1+\xi}+\Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon}
\end{aligned}$$

The marginal effect of taxes can then be summed up by

$$\frac{dv_{1,t}^{i}}{d\tau_t}=\left(\hat{c}_{1,t}^{i}\right)^{1-1/\rho}\frac{d\ln\left(\hat{c}_{1,t}^{i}\right)}{d\tau_t} \tag{B.8}$$

$$\begin{aligned}
\frac{d\ln\left(\hat{c}_{1,t}^{i}\right)}{d\tau_t} =&\frac{1+\xi}{\xi}\frac{d\ln\left(h_t\right)}{d\tau_t}+\frac{B_{t+1}}{1+B_{t+1}}(1-\alpha)\left(\frac{d\ln\left(h_{t+1}\right)}{d\ln\left(s_t\right)}-1\right)\frac{d\ln\left(s_t\right)}{d\tau_t}\\
&+\frac{\Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{1-\theta}{1+\gamma_0\epsilon}\left(\frac{d\tau_{t+1}}{d\tau_t}+\tau_{t+1}\frac{d\ln(\Gamma_{s,t})}{d\tau_t}\right)}{\frac{\eta_i^{1+\xi}}{X_i^{\xi}}\frac{1}{1+\xi}+\Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon}}
\end{aligned} \tag{B.9}$$

For the young informal households, we can employ the Euler equation in a similar way and write:

$$\begin{aligned}
v_{1,t}^{0} =&\frac{\left(\tilde{c}_{1,t}^{0}(1+B_{t+1}^{0})^{\frac{\rho}{\rho-1}}\right)^{1-1/\rho}}{1-1/\rho}=\frac{\left(\hat{c}_{1,t}^{0}\right)^{1-1/\rho}}{1-1/\rho}\\
\hat{c}_{1,t}^{0} =&\left(\frac{s_{t-1}}{\nu_t}\right)^{\sigma_s}(1+B_{t+1}^{0})^{\frac{1}{\rho-1}}\mathcal{C}_{1,t}^{0}\\
\mathcal{C}_{1,t}^{0} =&\frac{\eta_0^{1+\xi}}{X_0^{\xi}}\frac{1}{1+\xi}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}+\Theta_{s,t}\frac{1-\alpha}{r_0\alpha}\frac{\tau_{t+1}\epsilon}{1+\gamma_0\epsilon}
\end{aligned}$$

We then have that:

$$\frac{dv_{1,t}^{0}}{d\tau_t}=\left(\hat{c}_{1,t}^{0}\right)^{1-1/\rho}\frac{d\ln\left(\hat{c}_{1,t}^{0}\right)}{d\tau_t} \tag{B.10}$$

$$\frac{d\ln\left(\hat{c}_{1,t}^{0}\right)}{d\tau_t}=\frac{B_{t+1}^{0}}{1+B_{t+1}^{0}}(1-\alpha)\left(\frac{d\ln\left(h_{t+1}\right)}{d\ln\left(s_t\right)}-1\right)\frac{d\ln\left(s_t\right)}{d\tau_t}+\frac{\Theta_{s,t}\frac{1-\alpha}{r_0\alpha}\frac{\epsilon}{1+\gamma_0\epsilon}\left(\frac{d\tau_{t+1}}{d\tau_t}+\tau_{t+1}\frac{d\ln(\Theta_{s,t})}{d\tau_t}\right)}{\frac{\eta_0^{1+\xi}}{X_0^{\xi}}\frac{1}{1+\xi}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}+\Theta_{s,t}\frac{1-\alpha}{r_0\alpha}\frac{\tau_{t+1}\epsilon}{1+\gamma_0\epsilon}} \tag{B.11}$$

C Endogenous pension design

This appendix collects the derivations behind section 7: the corner that costless redistribution produces, the deadweight cost of redistribution and the condition under which an electorate chooses the Bismarckian corner, what the cost changes in the equilibrium of appendices A and B, and the state independence of the design choice under log preferences. Throughout there are no informal households, $\gamma_0 = 0$, so the budget (8) gives $\bar{b}_t = \nu_t w_t h_t \tau_t / h_{t-1}$ and a retiree of type i the benefit $\nu_t w_t h_t \tau_t \left(1 + \theta_t \left[y_i - 1\right]\right)$, where $y_i \equiv \eta_i^{1+\xi} / (X_i^{\xi} \Gamma_h) = \eta_i h_{t-1,i} / h_{t-1}$ is the labor income of type i relative to the mean by (A.3), and $\sum_i \gamma_i y_i = 1$.

C.1 Costless redistribution

We consider three timings of the choice. If θ_t is chosen at t together with τ_t, it re-splits benefits that current retirees have already earned. Current labor supply and savings depend on the design that will apply to the benefits they earn, θ_{t+1}, so the choice moves no economic decision, and its marginal effect on the political objective (14) runs entirely through current retirees,

$$\frac{\partial \mathcal{W}_t}{\partial \theta_t} = \omega \sum_i \gamma_i \mu_i \frac{\frac{1-\alpha}{\alpha} \tau_t \left(y_i - 1\right)}{\frac{s_{t-1}^i}{s_{t-1}} + \frac{1-\alpha}{\alpha} \tau_t \left(1 + \theta_t \left[y_i - 1\right]\right)}. \tag{C.1}$$

The numerator is the change in the benefit of type i and averages to zero across types: a more Bismarckian split moves a given total from retirees below mean income to retirees above it. The denominator is proportional to the consumption of a retiree of type i and rises with income. With equal propensities to vote the sum therefore weights the gains of the rich by a lower marginal utility than the losses of the poor. The following proposition makes the argument exact.

Proposition 3. *Assume log-GHH preferences, no informal households and no deadweight cost of redistribution (section 7.1), and let the design θ_t that applies to the benefits paid at t be chosen at t together with τ_t. Let the propensities to vote be positive and not rise with relative income, $\mu_i \leq \mu_j$ whenever $y_i > y_j$, equal propensities included, and let relative incomes not all be equal. Then, for every $\omega > 0$ and $\tau_t > 0$, the derivative (C.1) is negative at every $\theta_t \in [0, 1]$ at which every retiree's consumption is positive, and the design chosen together with a positive tax is the Beveridgean corner, $\theta_t = 0$. This holds for all values of α, β, ξ, ν_t, γ_i, η_i and X_i, and whatever tax and design current retirees expected when they saved.*

Proof of proposition 3. By (11) the labor supply and the savings of the young at t depend on the design that will apply to their own benefits, θ_{t+1}, and not on θ_t, so θ_t

enters no state at $t+1$ and no later decision. It enters $\mathcal{W}_t$ only through the consumption of current retirees (appendix A.1, with $\gamma_0 = 0$),

$$c_{2,t}^i = \alpha \nu_t h_t^{1-\alpha} \left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha} D_i, \qquad D_i \equiv \frac{s_{t-1}^i}{s_{t-1}} + \frac{1-\alpha}{\alpha} \tau_t \left(1 + \theta_t \left[y_i - 1\right]\right), \tag{C.2}$$

where the first factor is common to all types and independent of θ_t. From (14),

$$\frac{\partial \mathcal{W}_t}{\partial \theta_t} = \omega \sum_i \gamma_i \mu_i \frac{\partial \ln c_{2,t}^i}{\partial \theta_t} = \omega \, \frac{1-\alpha}{\alpha} \, \tau_t \sum_i \gamma_i \left(y_i - 1\right) \frac{\mu_i}{D_i},$$

which is (C.1). We sign the sum in three steps.

First, relative incomes average to one. By the definition of Γ_h in (A.3), $\sum_i \gamma_i y_i = 1$, and since $\sum_i \gamma_i = 1$ (section 2.1), $\sum_i \gamma_i \left(y_i - 1\right) = 0$: the design moves a given total between retirees below and above mean income.

Second, D_i rises with y_i. The savings of current retirees were chosen at $t-1$. By part iii of proposition 1, that is (A.8) with $B_{t+1} = \beta$, in which the tax and the design are those that apply to the benefits of the generation that saves,

$$\frac{s_{t-1}^i}{s_{t-1}} = y_i + \frac{1-\alpha}{\alpha} \, \frac{\tau_t^e \left(1 - \theta_t^e\right)}{1+\beta} \left(y_i - 1\right),$$

where τ_t^e and θ_t^e are the tax and the design that generation $t-1$ expected for period t. Whatever these were, the slope in y_i is at least one, since $\tau_t^e \geq 0$ and $\theta_t^e \leq 1$. The second term of D_i has slope $\frac{1-\alpha}{\alpha} \tau_t \theta_t \geq 0$, so D_i rises with y_i with a slope of at least one, and by (C.2) it is positive wherever every retiree's consumption is.

Third, Chebyshev's sum inequality: for weights $\gamma_i \geq 0$ that sum to one,

$$\sum_i \gamma_i a_i b_i - \sum_i \gamma_i a_i \sum_j \gamma_j b_j = \frac{1}{2} \sum_{i,j} \gamma_i \gamma_j \left(a_i - a_j\right)\left(b_i - b_j\right),$$

which is non-positive when a_i and b_i are ordered in opposite directions. Let $a_i = y_i - 1$ and $b_i = \mu_i / D_i$. If $y_i > y_j$, then $\mu_i \leq \mu_j$ and $D_i > D_j > 0$, so $b_i < b_j$, and every term with $y_i \neq y_j$ is negative. Since $\sum_i \gamma_i a_i = 0$ by the first step and relative incomes are not all equal, $\sum_i \gamma_i \left(y_i - 1\right) \mu_i / D_i < 0$, and so is the derivative.

Finally, D_i falls with θ_t only for types below mean income, and $D_i \geq 1$ for the others. The designs at which every retiree's consumption is positive therefore form an interval that starts at $\theta_t = 0$ whenever there are any, $\mathcal{W}_t$ is strictly decreasing on it, and the design chosen with any positive tax is $\theta_t = 0$.

The proof uses the propensities to vote only through the ratios μ_i / D_i, which order the types as $\mu_i / c_{2,t}^i$ does. The derivative is therefore negative at every design at which propensities to vote rise with relative income, if at all, by proportionally less than consumption in retirement, $\mu_i / \mu_j < c_{2,t}^i / c_{2,t}^j$ whenever $y_i > y_j$, and a design other than the

Beveridgean corner requires propensities to vote that, for some pair of types and at some design, rise at least as fast as consumption. The argument also holds under CRRA-GHH preferences at every intertemporal elasticity: the design still enters only the consumption of current retirees, whose marginal utility is $D_i^{-1/\rho}$ times a factor common to all types, and relative savings keep the form of (A.8) with a discount term B_t common to all types, so $\mu_i D_i^{-1/\rho}$ takes the place of μ_i/D_i.

Propensities to vote that rise with income can overturn the corner, and they do rise in the U.S. calibration of section 6, so there the sign is a quantitative matter: evaluated at the calibrated 2020 equilibrium, (C.1) is negative over the whole interval, at every date of the projection and at $\rho = 0.5$, 1 and 2, and the choice is the corner $\theta_t = 0$.

If instead θ_{t+1} is chosen at t, the design no longer touches the benefits of current retirees but moves current labor supply and savings, directly and through the anticipated response of τ_{t+1}. With the political objective (14) the marginal effect is

$$\frac{\partial \mathcal{W}_t}{\partial \theta_{t+1}} = \sum_i \gamma_i \mu_i \left[\omega \left(1-\alpha\right) \frac{d \ln \Theta_{h,t}}{d\theta_{t+1}} + \nu_t \left((1+\beta) \frac{d \ln \tilde{c}^i_{1,t}}{d\theta_{t+1}} + \beta \frac{d \ln R_{t+1}}{d\theta_{t+1}} \right) \right], \tag{C.3}$$

where $\Theta_{h,t}$ is the labor supply function (A.6) and each total derivative includes the response of τ_{t+1}, and of θ_{t+2}, to the design chosen. Retirees are unanimous, since their term does not depend on i, and they favour a more earnings-related system: a higher θ_{t+1} raises the return to an hour worked at t and with it the labor supply on which their benefits are levied. The young are divided between this forward-looking stake and the redistribution of their own future benefits, the force of (C.1) one period ahead. The equilibrium is a sequence of policy functions over the inherited design, identified by backward induction as in section 4, and at observed U.S. inequality the redistributive force dominates: the choice is again the corner $\theta = 0$. Section 7 adopts this timing. Of the three, it is the one that keeps the Markov structure and lets the design move labor supply and savings before it binds: a design chosen with the tax moves no decision, and a design chosen once and for all requires a commitment that the Markov equilibria of the paper do not have. It also mirrors how reforms of benefit formulas are legislated, applying to future benefits and leaving benefits in payment untouched.

A design chosen once and for all enters every channel at once. The joint choice of τ_{t_0} and θ is nevertheless one-dimensional, since at any candidate design the first order condition for the tax is the tax condition of appendix B.1 at that design, and the predetermined distribution of savings is held at the value the chosen design implies, which makes the equilibrium a fixed point. The outcome is again a corner: $\theta = 0$ under log preferences, and under CRRA preferences the Bismarckian corner $\theta = 1$ at high intertemporal elasticities, where the forward-looking channels are strong. Near the elasticity at which the corner switches the electorate is close to indifferent between the two, so the switch is not sharply located. The online appendix reports the three timings at each intertemporal

elasticity (online appendix); the technical documentation gives the algorithms (Berg and Gonzalez-Eiras 2026).

C.2 The deadweight cost of redistribution

The implicit tax. A retiree of type i receives $\nu_t w_t h_t \tau_t \left[\theta_t y_i + 1 - \theta_t\right]$ on past earnings $w_{t-1}\eta_i h_{t-1,i} = w_{t-1} y_i h_{t-1}$, that is

$$\Pi_t \, \tau_t \left[\theta_t + \frac{1-\theta_t}{y_i}\right] \text{ per unit of past earnings,} \qquad \Pi_t \equiv \frac{\nu_t w_t h_t}{w_{t-1} h_{t-1}},$$

where Π_t is the pay-as-you-go return factor: a purely earnings-related system paying the same revenue returns $\Pi_t \tau_t$ per unit of earnings to every type. The flat component moves type i away from that benchmark by

$$t_i \equiv \Pi_t \, \tau_t \, (1-\theta_t) \left(1 - \frac{1}{y_i}\right) \tag{C.4}$$

per unit of earnings: a tax on types above mean income, a subsidy to types below it, and a pure transfer in the aggregate, since $\sum_i \gamma_i y_i t_i = 0$. This is the tax component of the contribution in the sense of Summers (1989), the part of the payroll charge that buys the contributor no benefit, and the only part that distorts like a tax.

Which margin. The wedge (C.4) is an average implicit tax: it compares what type i receives per unit of its earnings with the benchmark. The marginal wedge is a different object. The flat component depends on aggregate hours and not on own hours, so an extra hour raises the benefit of every type by $\Pi_t \tau_t \theta_t$ per unit of earnings, and the marginal wedge of the flat component, $\Pi_t \tau_t (1-\theta_t)$, is common to all types. That wedge is already in the model: it is the group $\frac{1-\alpha}{\alpha}\theta \tau_{t+1} \Gamma_{s,t}$ in the labor supply function (A.6), through which hours respond to the design with the Frisch elasticity ξ, and GHH preferences carry no income effect through which the transfers between types could move hours. A cost built on the hours margin would count that response twice and could not depend on the spread of incomes. The cost therefore stands for margins the model does not contain, participation, formality and the reporting of earnings, on which the average wedge is the relevant one: a worker who decides whether to bring earnings into the system compares the earnings at stake with what they buy, so the wedge that matters is the average tax on those earnings (Saez 2002; Kleven and Kreiner 2006).

The Harberger loss. A set of proportional wedges t_i on bases e_i, each responding to its net-of-tax rate with a common elasticity ε, costs $\frac{1}{2}\varepsilon \sum_i \gamma_i t_i^2 e_i$ to second order in the wedges (Harberger 1964). Per unit of mean past earnings the bases are $e_i = y_i$, so the

loss is $\frac{1}{2}\varepsilon\,\Pi_t^2\tau_t^2(1-\theta_t)^2\tilde{V}$ with

$$\tilde{V} \equiv \sum_i \gamma_i \frac{(y_i-1)^2}{y_i}, \tag{C.5}$$

while revenue per unit of mean past earnings is $\Pi_t\tau_t$. As a share of revenue the loss is therefore $\frac{1}{2}\varepsilon\,\Pi_t\,\tau_t\,(1-\theta_t)^2\,\tilde{V}$: quadratic in the flat share of the system, linear in its size and proportional to a dispersion of relative incomes that is zero when all incomes are equal.

Specification. The exponential form $f(\theta_t,\tau_t)$ of (15) scales both components of the benefit alike,

$$A(\theta_t,\tau_t) = f(\theta_t,\tau_t)\,\theta_t, \qquad B(\theta_t,\tau_t) = f(\theta_t,\tau_t)\,(1-\theta_t), \tag{C.6}$$

so that the surviving share of revenue is $A+B=f\in(0,1]$. We let the leak of Okun's bucket (Okun 1975) have two parts: a constant share of every unit the flat component carries, and a share that rises with the load. The first needs the amount carried. The implicit tax (C.4) moves $y_i|t_i| = \Pi_t\tau_t(1-\theta_t)|y_i-1|$ per unit of mean past earnings to or from a retiree of type i, and since what is taken above mean income is what arrives below it, the sum over types counts every unit twice: the flat component carries

$$\tfrac{1}{2}\sum_i \gamma_i y_i\,|t_i| = \tfrac{1}{2}\,\Pi_t\,\tau_t\,(1-\theta_t)\,M, \qquad M \equiv \sum_i \gamma_i\,|y_i-1|\,, \tag{C.7}$$

with M the mean absolute deviation of relative income. A leak that loses a constant share of every unit carried therefore costs a share of revenue proportional to $(1-\theta_t)M$, while the Harberger loss above is a share proportional to $\tau_t(1-\theta_t)^2\tilde{V}$. We write

$$f(\theta_t,\tau_t) = \exp\Big\{-\tau_t\,(1-\theta_t)\Big[\lambda_1 M + \tfrac{1}{2}\lambda_2\,\tilde{V}\,(1-\theta_t)\Big]\Big\}, \tag{C.8}$$

so that $1-f$ is, to first order, $\lambda_1\tau_t M(1-\theta_t) + \frac{1}{2}\lambda_2\tau_t\tilde{V}(1-\theta_t)^2$, with λ_2 in the place of $\varepsilon\Pi_t$.[28] Each term carries the dispersion its own derivation gives: a leak proportional to the amount carried counts every unit alike, so its weight is M; a loss quadratic in each type's wedge counts a unit by the wedge it travels under, $|y_i-1|/y_i$, so that a given transfer counts for more against a smaller income, which is $\tilde{V}$. The one choice in (C.8) is the factor τ_t on the linear term: a constant share of the amount carried is, as a share of revenue, independent of the size of the system, and we let it rise with the size of the system as the quadratic share does.[29] The quadratic cost is $\lambda_1 = 0$ and the linear cost

[28] The return factor multiplies every implicit tax alike and so scales the loss once. It carries ν_t and would make a structural cost parameter drift with demographics; we absorb it into the cost parameters, which are calibrated rather than derived.

[29] Under the literal leak, $1-f \approx \lambda_1(1-\theta_t)M + \frac{1}{2}\lambda_2\tau_t\tilde{V}(1-\theta_t)^2$, the cost parameter at which an

$\lambda_2 = 0$; the general case is written as a scale and a shape, $\lambda_1 = \lambda\kappa$ and $\lambda_2 = \lambda$. The exponential keeps f positive for every parameter and every size of system, and nests the model without the cost at $\lambda = 0$. The three properties of section 7.1 hold for every variant: $f = 1$ at $\theta_t = 1$ and, for every design, when all incomes are equal, where M and $\tilde{V}$ vanish together; the share lost rises with τ_t, and since $\ln f$ is linear in τ_t the marginal contribution is worth $1 + \ln f$ of its revenue to a retiree whatever the variant (appendix C.3); and the marginal cost of the first unit of redistribution is

$$\left.\frac{\partial \ln f}{\partial \theta_t}\right|_{\theta_t=1} = \lambda_1 \tau_t M,$$

zero under the quadratic cost and positive under the two with a linear term. Scaling both components alike is a choice. A cost on the flat component alone delivers the same endpoints and differs in the interior, but it loses the separate identification of θ and the cost parameters (appendix C.3). A cost on the design itself rather than on the transfer it performs, $f(\theta) = \phi + (1 - \phi)\theta^p$, stays in place when a compressed income distribution removes the redistributive stakes; the online appendix reports it.

The Bismarckian corner. With the design chosen one period in advance under log preferences the choice maximizes a function $\mathcal{B}_t(\theta_{t+1})$ of the candidate design alone (appendix C.4), and at $\theta_{t+1} = 1$ the cost enters its slope through one number, the share of revenue the first unit of design saves, $\ell_{t+1} \equiv \lambda_1 \tau_{t+1} M$, with $\tau_{t+1} = \mathcal{T}_{t+1}(1)$ the tax the young anticipate at the Bismarckian design; λ_2 does not enter. At a given response of that tax to the design the slope is affine and increasing in ℓ_{t+1}, so the corner is a local maximum of $\mathcal{B}_t$ once λ_1 exceeds $\underline{\lambda}_1 = \bar{\ell}_{t+1}/(\tau_{t+1} M)$, where $\bar{\ell}_{t+1}$, the value of ℓ_{t+1} at which the slope vanishes, is the obstacle to the corner: the loss of the flat benefit to the young, net of what they and the retirees gain in hours and returns. The technical documentation derives the slope in closed form (Berg and Gonzalez-Eiras 2026); three of its properties carry section 7. Every term of the slope is of order τ_{t+1}, since the young's stake in the design and the weight of the pension in their lifetime resources are both proportional to the tax, so $\bar{\ell}_{t+1}$ is independent of the size of the system to first order and $\underline{\lambda}_1$ is proportional to $1/\tau_{t+1}$. A flatter voting profile raises the obstacle, since the loss of the flat benefit weighs each type by the inverse of its relative income and a profile that rises with income, as in the U.S., weighs the poor less. And ageing lowers it, since a scarcer young generation raises the weight of retirees, who are unanimous for a more earnings-related system, and raises the tax the young anticipate.The income distribution enters on both sides, through

electorate chooses the Bismarckian corner is independent of the size of the system to first order, so the corner is as far for a large system as for a small one, and only the quadratic part dissipates at the margin of the tax, so that the factor of appendix C.3 becomes $1 + \ln f + \lambda_1(1 - \theta_t)M$. With the factor τ_t that parameter is proportional to $1/\tau_{t+1}$, as the corner condition below shows, which is what lets the size of the system separate France from the UK at nearly the same dispersion, and one factor $1 + \ln f$ serves every variant.

the amount the flat component carries, which falls with it, and through the loss of the flat benefit and the tax response, which fall with it too; at a given tax response a more compressed distribution raises the obstacle, since the amount carried falls by less than the redistributive part of the loss, but once the tax response is included the net effect at the U.S. calibration is the one section 7.3 reports, a compressed distribution moving the chosen design toward the corner. The tax response is itself a function of the cost: without the cost a more Bismarckian design lowers retirees' support for the tax, as in proposition 2, and the cost raises the response with ℓ_{t+1}, since the dissipation the design saves raises what the marginal contribution is worth to retirees; at the three calibrated economies it is positive at the threshold. The thresholds of section 7.2 are computed from the choice, as the smallest parameter at which the electorate chooses the corner, which is the root of the slope where $\mathcal{B}_t$ is single-peaked, as the technical documentation verifies at the three economies.

Magnitude. The shape of f follows from the tax component; its size does not. The quadratic parameter stands where $\varepsilon\Pi_t$ does in the Harberger share: with Π_t of the order of ν_t, 1.34 in the U.S. calibration for 2020 (table 3), participation and taxable-income elasticities of 0.1 to 0.4 (Chetty et al. 2011; Saez et al. 2012) correspond to λ_2 between about 0.13 and 0.54, while the values calibrated in section 7 are an order of magnitude above that range at $\rho \leq 1$, and under the quadratic cost three to thirteen times above it at $\rho = 2$, so λ_2 also stands in for administrative, evasion and political costs of redistribution that the model does not contain. The linear parameter has no elasticity benchmark: it sets the share of each unit carried that is lost whatever the wedge it travels under, $2\lambda_1\tau_t$ by (C.7), the leak that administrative and political costs of redistribution absorb, and section 7.2 pins it by a second country's design rather than by an elasticity.

C.3 The equilibrium with the cost

With the cost, the benefit formula of (8) becomes

$$b_t^i = \left[A(\theta_t, \tau_t)\, h_{t-1,i}\eta_i + B(\theta_t, \tau_t)\, h_{t-1}\right] \bar{b}_t, \tag{C.9}$$

with $\bar{b}_t = \nu_t w_t h_t \tau_t / h_{t-1}$ the gross revenue per unit of past labor, of which a share $1 - f$ does not reach retirees. The model without the cost is $(A, B) = (\theta_t, 1 - \theta_t)$.

Economic equilibrium. In appendix A the design enters only through the groups $\frac{1-\alpha}{\alpha}\tau_{t+1}\theta_{t+1}$ and $\frac{1-\alpha}{\alpha}\tau_{t+1}(1 - \theta_{t+1})$, in $\Gamma_{s,t}$, $\Theta_{h,t}$ and $s_{t,i}/s_t$, and through θ_t and $1 - \theta_t$ in the benefits of current retirees. The cost is therefore exactly the substitution $\theta_{t+1} \mapsto A(\theta_{t+1}, \tau_{t+1})$ and $1 - \theta_{t+1} \mapsto B(\theta_{t+1}, \tau_{t+1})$ in those groups, and $\theta_t \mapsto A(\theta_t, \tau_t)$, $1 - \theta_t \mapsto B(\theta_t, \tau_t)$ in the benefit formula, and it changes nothing else. Under log preferences, (A.5),

(A.6) and (A.8) read

$$\Gamma_{s,t} = \frac{1}{1+\xi}\,\frac{\Gamma_h \beta}{1+\beta+\frac{1-\alpha}{\alpha}\tau_{t+1}\left[A(\theta_{t+1},\tau_{t+1})(1+\beta)+B(\theta_{t+1},\tau_{t+1})\right]}, \tag{C.10a}$$

$$\Theta_{h,t} = (\Gamma_h)^{\frac{1+\xi}{1+\alpha\xi}}\left(\frac{(1-\alpha)(1-\tau_t)}{\Gamma_h - \frac{1-\alpha}{\alpha}A(\theta_{t+1},\tau_{t+1})\,\tau_{t+1}\Gamma_{s,t}}\right)^{\frac{\xi}{1+\alpha\xi}}, \tag{C.10b}$$

$$\frac{s_{t,i}}{s_t} = y_i + \frac{1-\alpha}{\alpha}\,\frac{B(\theta_{t+1},\tau_{t+1})\,\tau_{t+1}}{1+\beta}\,(y_i-1)\,, \tag{C.10c}$$

and the consumption of a retiree of type i carries $A(\theta_t,\tau_t)\,y_i + B(\theta_t,\tau_t)$ in place of $1+\theta_t\,[y_i-1]$ in (C.2). The two coincide when $A+B=1$, so the model without the cost is nested exactly. Here τ_{t+1} is the tax the young anticipate for the period in which their benefits are paid, $\tau_{t+1} = \mathcal{T}_{t+1}(\theta_{t+1})$ with $\mathcal{T}_{t+1}$ the equilibrium tax function below, so the young anticipate both the design and the size of the system that will pay them.

The tax condition. The first order condition for the tax of appendix B.1 picks up one factor. The benefit term of the retirees' marginal effect $\mathcal{E}^i_{2,t}$ is multiplied by τ_t, and f now moves with τ_t, so

$$\mathcal{E}^i_{2,t} = (1-\alpha)\frac{\partial \ln(\Theta_{h,t})}{\partial \tau_t} + \frac{\frac{1-\alpha}{\alpha}\left[A(\theta_t,\tau_t)\,y_i + B(\theta_t,\tau_t)\right]\left(1+\tau_t\frac{\partial \ln f(\theta_t,\tau_t)}{\partial \tau_t}\right)}{\frac{s^i_{t-1}}{s_{t-1}} + \frac{1-\alpha}{\alpha}\tau_t\left[A(\theta_t,\tau_t)\,y_i + B(\theta_t,\tau_t)\right]}, \tag{C.11}$$

where $\partial \ln f/\partial\tau_t = \ln f/\tau_t$ for every variant of (C.8), since $\ln f$ is linear in τ_t, and hence $1+\tau_t\,\partial \ln f/\partial\tau_t = 1+\ln f(\theta_t,\tau_t)$ whatever the variant: a larger system is worth less to a retiree at the margin than the revenue it collects, by the share of the marginal contribution that the flat component dissipates. Nothing else in the condition sees f. The term of the young, $\mathcal{E}^i_{1,t}$, contains no policy but τ_t, and $\partial \ln(\Theta_{h,t})/\partial\tau_t = -\frac{\xi}{1+\alpha\xi}\frac{1}{1-\tau_t}$ is free of the groups, so the condition involves θ_{t+1} nowhere and the equilibrium tax at t is a function $\mathcal{T}_t(\theta_t)$ of the inherited design and of parameters and demographics alone, as in proposition 2.

Identification of θ. The replacement rate of type j, its benefit over its past earnings, is $\Pi_t\tau_t\,[A+B/y_j]$. The ratio of the replacement rates of two types, the datum that identifies θ in section 6, therefore depends on (A,B) only through $B/A = (1-\theta_t)/\theta_t$, from which f cancels: the design is identified separately from the cost, and $\theta^* = 0.738$ for the U.S. whatever λ_1 and λ_2. This is a property of the proportional specification. A cost on the flat component alone, $(A,B) = (\theta_t, f(1-\theta_t))$, leaves $B/A = f(1-\theta_t)/\theta_t$, so that the same datum implies a different θ for every value of the cost parameter and the two are identified jointly.

The design condition. With the design chosen one period in advance, θ_{t+1} enters the objective through $A(\theta_{t+1},\tau_{t+1})$ and $B(\theta_{t+1},\tau_{t+1})$ in (C.10), with τ_{t+1} the anticipated

equilibrium tax. At a given tax, $\partial A/\partial\theta_{t+1} = f + \theta_{t+1}\,\partial f/\partial\theta_{t+1}$ and $\partial B/\partial\theta_{t+1} = (1-\theta_{t+1})\,\partial f/\partial\theta_{t+1} - f$ with $\partial f/\partial\theta_{t+1} = f\tau_{t+1}\left[\lambda_1 M + \lambda_2\tilde{V}(1-\theta_{t+1})\right]$ by (C.8), so

$$\frac{\partial A}{\partial\theta_{t+1}} + \frac{\partial B}{\partial\theta_{t+1}} = \frac{\partial f}{\partial\theta_{t+1}} = f\,\tau_{t+1}\left[\lambda_1 M + \lambda_2\,\tilde{V}\,(1-\theta_{t+1})\right] \geq 0, \tag{C.12}$$

rising with the dispersion of incomes and the size of the system the young anticipate, and with the flat share through the quadratic term; at the corner it is zero under the quadratic cost and $\lambda_1\tau_{t+1}M$ otherwise. Without the cost a more Bismarckian design shifts next period's benefits one for one from the flat to the earnings-related component, $\partial A/\partial\theta_{t+1} = -\partial B/\partial\theta_{t+1} = 1$, and the redistributive stake of the young dominates as in appendix C.1. With it, a more Bismarckian design also raises the total the young will receive, since it dissipates less, which is what lets the design condition cross zero in the interior and, when the linear term is large enough, stay positive up to the Bismarckian corner (the corner condition of appendix C.2).

C.4 The design choice under log preferences

Under log preferences, with no informal households and the design chosen one period in advance under the cost (C.6), the political objective at t is additively separable,

$$\mathcal{W}_t = \mathcal{A}_t\,(\tau_t, \theta_t) + \mathcal{B}_t\,(\theta_{t+1}) + \mathcal{C}_t\,(s_{t-1})\,, \tag{C.13}$$

where $\mathcal{C}_t$ depends on no policy: $\Theta_{h,t}$ in (C.10b) is the product of a factor in τ_t and a factor in $(\tau_{t+1}, \theta_{t+1})$, with $\tau_{t+1} = \mathcal{T}_{t+1}(\theta_{t+1})$ by the tax condition, so every dependence of the young's utility on θ_{t+1} arrives additively, while the consumption of current retirees carries (τ_t, θ_t) and s_{t-1} alone. The tax chosen at t therefore depends on the inherited design alone, and the design chosen at t maximizes $\mathcal{B}_t$ and depends on neither θ_t nor s_{t-1}; the proof is in the technical documentation (Berg and Gonzalez-Eiras 2026). This is what section 4 uses: each date's choice is a search over a grid of designs with no state, the tax at each candidate design is found as in proposition 2, and the condition $\theta_{2020} = \theta^*$ that calibrates λ in section 7.2 is a function of λ alone. Under CRRA preferences the powers do not separate, both s_{t-1} and θ_t re-enter the design choice, and the equilibrium is a backward recursion over policy functions of the state (s_{t-1}, θ_t).

D Terminal states

We consider two different terminal states of the model: A finite horizon version where the economy ends after $t = T$ and a steady state version. The paper reports the finite-horizon solution, in periods of thirty years from 1960 to $T = 2200$ for the rich economies and

from 1920 to $T = 2190$ for Argentina, with population growth held at its last projected rate after 2080 (2070 for Argentina).

Finite Horizon. In the finite horizon, consumption functions and indirect utility in economic equilibrium are those of the general case in every period $t < T$. For the young generations at time T, we simply say that

$$u_{T,j} = u_j\left(c^j_{1,T}, h_{T,j}\right).$$

In this case, the economic equilibrium at $t = T$ is the general formulation with $\beta = \tau_{t+1} = s_t = s^j_t = 0$. Specifically:

$$h_t = (\Gamma_h)^{\frac{1}{1+\alpha\xi}} \left((1-\alpha)(1-\tau_t)\right)^{\frac{\xi}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha\xi}{1+\alpha\xi}} \tag{D.1}$$

$$h_{t,i} = h_t \frac{(\eta_i/X_i)^\xi}{\Gamma_h} \tag{D.2}$$

$$\tilde{c}^i_{1,t} = \frac{1}{1+\xi}\frac{\eta_i^{1+\xi}}{X_i^\xi}\left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \tag{D.3}$$

$$\tilde{c}^0_{1,t} = \frac{1}{1+\xi}\frac{\eta_0^{1+\xi}}{X_0^\xi}\left(w^0_t\right)^{1+\xi}. \tag{D.4}$$

For the retired generation, the relevant consumption functions are unchanged.

For the politico-economic equilibrium, the marginal effect of τ_t on retirees is the same as for $t < T$. For working households, the effects are much simpler, namely:

$$\mathcal{E}^i_{1,t} = \left(\tilde{c}^i_{1,t}\right)^{1-1/\rho} \frac{1+\xi}{\xi}\frac{\partial \ln(h_t)}{\partial \tau_t} \tag{D.5}$$

$$\mathcal{E}^0_{1,t} = 0 \tag{D.6}$$

Steady state, infinite horizon. In steady state, we combine equations for aggregate labor supply and savings states to write:

$$s^* = \Gamma_h^{1+\xi}\left[\left(\frac{(1-\alpha)(1-\tau)}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau}{1+\gamma_0\epsilon}\Gamma_s}\right)^{1+\xi}\frac{\Gamma_s^{1+\alpha\xi}}{\nu^{\alpha(1+\xi)}}\right]^{\frac{1}{1-\alpha}} \tag{D.7}$$

$$\Gamma_s = \frac{1}{1+\xi}\frac{\Gamma_h B}{1+B+\frac{1-\alpha}{\alpha}\frac{\tau}{1+\gamma_0\epsilon}(1+\theta B)} \tag{D.8}$$

$$B = (\beta)^\rho (R^*)^{\rho-1}, \qquad R^* = \alpha\left(\frac{s^*}{\nu h^*}\right)^{\alpha-1}$$

$$\frac{s^*}{\nu h^*} = \frac{(s^*)^{\frac{1}{1+\xi}}\Gamma_s^{\frac{\xi}{1+\xi}}}{\nu\Gamma_h} = \left(\frac{(1-\alpha)(1-\tau)}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau}{1+\gamma_0\epsilon}\Gamma_s}\frac{\Gamma_s}{\nu}\right)^{\frac{1}{1-\alpha}} \tag{D.9}$$

We can use this to write the discount function as

$$B = (\beta)^{\rho} \left(\alpha \frac{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau}{1+\gamma_0\epsilon} \Gamma_s}{(1-\alpha)(1-\tau)} \frac{\nu}{\Gamma_s} \right)^{\rho-1} \tag{D.10}$$

E Calibration – Argentina

This appendix holds the data, the identifying formulas and the sources behind table 1, which section 5 summarizes.

E.1 The household survey

The Encuesta Permanente de Hogares (EPH) of the Instituto Nacional de Estadística y Censos (INDEC) is Argentina's main household survey of employment, hours and income, and the basis of its official labor market statistics. It is a continuous quarterly survey of the urban agglomerates, 31 metropolitan areas at present, in which households are drawn by a stratified multi-stage design. We use the last two quarters of 2006, a phase of sustained expansion after the recovery from the 2001–2002 crisis, at the start of the expansion of coverage under the amnesties (figure 5.1) and before the Great Recession. The survey asks respondents whether they contribute to social security, which separates formal from informal workers, and we keep those aged 25 to 60.

E.2 The pension system before and after the reform

Before the tax amnesties, retirees over the age of 70 who had made contributions for a minimum of 10 years were eligible for basic pension coverage. This coverage consisted of 70% of the minimum pension and a proportional amount from other contributions.[30] We calculate the pre-reform ϵ as

$$\epsilon = 0.7 \left(1 - \theta + \theta \frac{\eta_1 h_{t-1,1}}{h_{t-1}} \right) \frac{\beta^{5/30 \frac{\text{LE}_{\text{men}}-5}{\text{LE}_{\text{men}}}} + \beta^{10/30 \frac{\text{LE}_{\text{women}}-10}{\text{LE}_{\text{women}}}}}{2},$$

where $\text{LE}_{\text{men}} = 14.45$ and $\text{LE}_{\text{women}} = 22.55$ are the expected life expectancy for men and women, respectively, from their retirement ages (women retire at age 60, men at 65), such that the formula above corrects for the basic pension being received 5 or 10 years after the formal retirement age.[31] We set γ_0 so that informal households are 32% of all households,

[30] The basic pension was set by Law 24,347 of 1994 at 70% of the first pillar, denoted "prestación básica universal", plus the corresponding fully-funded or pay-as-you-go benefit. For simplicity we assume the latter on average was also 70% of the benefits received by workers that had at least 30 years of contributions. Thus, basic pensions were 70% of the benefits received by the least productive formal workers.

[31] To estimate life expectancy at retirement age for Argentina we conduct a regression analysis using data for men and women's life expectancies at birth and at age 65 for 40 OECD countries for the year

the share of the population over 65 without a pension before the reform, in 2004 (Rofman and Oliveri 2012); since the formal shares sum to one, $\gamma_0 = 0.32/0.68 = 0.47$. To identify the Bismarckian incentives, we use OECD data on the ratio of replacement rates, RR, at half mean and mean wages for males and females.[32] Specifically, we adjust θ to ensure that

$$RR \equiv \frac{\theta + (1-\theta)\frac{h_t}{\eta_3 h_{t,3}}}{\theta + (1-\theta)\frac{h_t}{\eta_1 h_{t,1}}} = \frac{0.678}{0.803},$$

where types 1 and 3, the first and the third quartiles, are those whose mean incomes are closest to half the formal average and to the average, at 0.47 and 1.00 of the average in the survey.

E.3 Technology, demographics, and the aggregate targets

The capital income share is that of the formal sector. In the model, informal households produce outside Y_t and K_t, and national accounts book the output of unincorporated household enterprises, where most informal work sits, as mixed income, which a share computed as one minus the wage bill over GDP counts entirely as capital (Gollin 2002). The economy-wide share for Argentina is 0.43 (Frankema 2010). If a share y of GDP is informal mixed income, the formal-sector share is $(0.43-y)/(1-y)$. Informal earnings of about 5% of GDP, as in the household survey, give 0.40, while shadow-economy estimates of 20–25% of GDP give 0.24 to 0.29. We take $\alpha = 0.35$, which also puts Argentina on the same footing as the Gollin-adjusted shares we use for the rich economies in section 6. We set the Frisch elasticity of labor supply to $\xi = 0.30$. To calibrate β, we target the capital-output ratio. A model period is thirty years and capital depreciates fully between periods, so the capital stock in place at t is the previous generation's savings, $K_t = s_{t-1}/\nu_t$, and with Cobb–Douglas production $K_t/Y_t = \alpha/R_t$ — a stock relative to one period, i.e. thirty years, of output. Thirty times that ratio is the capital-output ratio as the national accounts report it. In our model with $K_t/Y_t = \alpha/R_t$ this is essentially equivalent to targeting the thirty year interest rate, which for Argentina is not credibly measured. We target $K/Y = 3.23$, Argentina's ratio in 2010.[33] We calibrate β and ϵ

2000. We then use the fitted regression and data from the World Bank on male and female life expectancy at birth in Argentina in 2000.

[32] In a pure Bismarckian system the replacement rate would be the same at all wage levels, such that $RR = 1$, while in a pure Beveridgean it would double if we halve wages, implying $RR = 0.5$. Note that we combine men and women since we do not have data separately on the increase in coverage or government spending by gender.

[33] Capital stock and GDP at constant national prices (`rnna`, `rgdpna`) from the Penn World Table 11.0 (Feenstra et al. 2015), retrieved through FRED. We read it at the calibration year, where the tax rate, the replacement-rate ratio, the coverage share and the household survey are all measured, rather than averaging. Correcting for the shadow economy would raise the capital-output target, since the Penn World Table stock is economy-wide and the informal sector is labor-intensive. We keep the measured

numerically, since β depends on the politico-economic equilibrium path for savings and labor supply and ϵ depends on β.

To construct the time series for ν_t we follow Gonzalez-Eiras and Niepelt (2008) and use 30-year gross population growth rate and projections from census data. Finally, we use social security spending before the 2001 crisis of 7.1% of GDP,[34] to calibrate ω. Since the payroll tax falls on formal labor income, which is a share $1-\alpha$ of formal output, spending of 7.1% of GDP corresponds to an equivalent social security tax rate of $\tau_{2010} = 0.071/(1-\alpha) = 0.109$.

E.4 Productivity and hours

We use η_j and X_j to match the income distribution across formal households and the income and working hours of informal households. Specifically, let z_j^η and z_j^x denote income and hours worked of group j relative to the population-weighted average over the formal households, so that $\sum_{i>0}\gamma_i z_i^\eta = \sum_{i>0}\gamma_i z_i^x = 1$; informal households are measured against the same formal average. With labor supply $h_t^i/h_t = (\eta_i/X_i)^\xi / \sum_{j>0}\gamma_j \eta_j^{1+\xi}/X_j^\xi$ for formal households (cf. proposition 1) and $h_t^0 = (\eta_0 w_t^0/X_0)^\xi$ for informal households, where w_t^0 is the informal wage, we solve the system

$$
\begin{aligned}
z_i^\eta &= \frac{w_t(1-\tau_t)h_t^i\eta_i}{w_t(1-\tau_t)h_t} = \frac{\eta_i^{1+\xi}/X_i^\xi}{\sum_{j>0}\gamma_j\eta_j^{1+\xi}/X_j^\xi}, \qquad i>0, \\
z_i^x &= \frac{h_t^i}{\sum_{j>0}\gamma_j h_t^j} = \frac{(\eta_i/X_i)^\xi}{\sum_{j>0}\gamma_j(\eta_j/X_j)^\xi}, \qquad i>0, \\
z_0^\eta &= \frac{w_t^0 h_t^0 \eta_0}{w_t(1-\tau_t)h_t}, \qquad z_0^x = \frac{h_t^0}{\sum_{j>0}\gamma_j h_t^j}.
\end{aligned}
\tag{E.1}
$$

Because only relative income and hours are observed, both the level of efficiency units and the unit in which hours are measured are free. We spend the first on the normalization $\sum_{i>0}\gamma_i\eta_i^{1+\xi}/X_i^\xi = 1$, and take the taste for leisure to be common across the formal groups, $X_i = X$ for $i>0$. The formal income conditions then invert in closed form, $\eta_i = (z_i^\eta)^{1/(1+\xi)} X^{\xi/(1+\xi)}$, and they satisfy the normalization at any X, so the level of X is what fixes the unit in which hours are counted. We pin it down by requiring average formal hours in the baseline year, $\sum_{i>0}\gamma_i h_t^i$, to equal the average workweek of formal workers in the survey, 42.5 hours expressed as a share of discretionary time. Relative hours across the formal quartiles are then a prediction rather than a target, rising with

ratio, because no source measures an informal capital share and the stock also includes housing, which cuts the other way.

[34] Total public pension spending, national and provincial schemes, 7.0% and 7.2% of GDP in 2000 and 2001 (Cetrángolo and Grushka 2020, table 13). We use pre-crisis spending as it was relatively stable during that time, and after 2010 contributive pensions returned to their pre-crisis levels. Following the 2001 crisis, pension spending was reduced, to 5.2% of GDP in 2005, by only allowing minimum pensions to adjust for inflation.

relative income at elasticity $\xi/(1+\xi)$, and section 5 compares the prediction with the survey. The aggregate h_t weights hours by productivity and is not average formal hours; the two differ by the factor $\sum_{i>0}\gamma_i(\eta_i/X_i)^{\xi}$, which the informal hours condition in (E.1) carries. That condition and the informal income condition pin down η_0 and X_0 given the equilibrium h_t in the baseline year, and we solve for them jointly with β and ω at $X = 1$. The level of X then follows in one step from the solved hours, and η_0 and X_0 are re-derived at it. That last step leaves the equilibrium untouched, since the informal household reaches every aggregate through $\eta_0^{1+\xi}/X_0^{\xi}$, which the unit of hours does not move.

E.5 The vector-X_i calibration

The alternative to a common taste for leisure treats the survey's relative hours as data at every point of the formal distribution: with both z_i^{η} and z_i^{x} imposed, (E.1) identifies one X_i per quartile, and the level of hours is then not identified. Since the productivity distribution reaches every aggregate only through $\sum_{i>0}\gamma_i\eta_i^{1+\xi}/X_i^{\xi}$, which both calibrations normalize to one, β, ω and every equilibrium outcome of section 5, the reform at every ρ and its dependence on ϵ and θ included, are identical under the two; only the productivity and leisure parameters and the reading of relative hours differ. That calibration, and the reform at every ρ under both, are reported in the online appendix.

F Calibration – the U.S., France, and the UK

This appendix documents the data and the household heterogeneity behind the calibrations of section 6, whose parameters are in table 3. Each country has three income groups of men aged 25 to 60. The tables report, group by group, the population share γ_i, the taste for leisure X_i and the productivity η_i that the calibration assigns, and the propensity to vote μ_i taken from the data. Under the calibration of section 6 the taste for leisure is common across groups, so the X_i row repeats one number, fixed by the country's average workweek, and the income distribution is carried by η_i alone; the online appendix describes the alternative, with one X_i per group identified from relative hours.

The parameters in the lower panel of table 3 are common to the three countries. The discount factor β matches the U.S. thirty-year interest rate, the gross interest rate net of average multifactor productivity growth estimated by Gonzalez-Eiras and Niepelt (2008), and is imposed on France and the UK. The Frisch elasticity is $\xi = 0.30$, as for Argentina, and the capital share $\alpha = 0.30$.[35] Pension spending is for 2020 (OECD 2021), and ν_t is thirty-year gross population growth from census data and projections,

[35] Averaging adjustments 2 and 3 of Gollin (2002) gives capital income shares close to 0.3 for France and the U.S. and slightly lower for the UK.

corrected for France as described below; the online appendix reports the calibration at every intertemporal elasticity of the model's grid.

F.1 The U.S.

For the U.S. we use the CPS March Supplement 2019. The low and medium income groups are cut so that their mean incomes are half the mean and the mean, which is what identifies θ from the replacement rates at those two income levels reported in OECD (2021); the remaining households form the high income group. The propensity to vote is from the U.S. Census, Voting and Registration in the Election of November 2020. To pension spending we add Medicare spending from the 2020 Data Book of the Medicare Payment Advisory Commission, and Medicare enters the calibration of θ as a benefit unrelated to income. France and the UK have no counterpart, since their public health spending is not confined to retirees.[36]

Table 6 shows the two features that set the U.S. apart. Productivity rises steeply across the groups, from 0.82 in the low group to 2.52 in the high one, and the propensity to vote rises with income, from 0.47 to 0.77, so the political objective weighs high-income households well above their population share. The low group holds 58% of households.

Table 6: Household heterogeneity – the U.S.

	Income group			
Parameter	**Low**	**Medium**	**High**	**Target**
γ_i	0.58	0.18	0.24	Income percentiles.
X_i	4.03	4.03	4.03	Average hours worked.
η_i	0.82	1.38	2.52	Income distribution.
μ_i	0.47	0.63	0.77	Voting propensity.

F.2 France

For France we use the 2018 wave of the Luxembourg Income Study. France's system is fully Bismarckian, so $\theta = 1$ whatever the income cuts, and the groups need not be cut at half-mean and mean income. We cut them at the U.S. percentiles instead, which puts the two countries' household types in one-to-one correspondence and makes the counterfactuals of section 6 a type-by-type replacement. The propensity to vote is from the Comparative Study of Electoral Systems.[37] Since the retirement age in France in

[36] Bethencourt and Galasso (2008) treat public health care and pensions as political complements, which supports counting Medicare among the transfers to the old that the U.S. electorate votes on.

[37] https://cses.org/.

2020 was 63.5 years, against 66 in the U.S. and the UK, we take France's population growth in 2020 to be

$$\frac{\text{Population 2020} \times (30 - 2.5)/30}{\text{Population 1990} + \text{Population 2020} \times 2.5/30},$$

and adjust later years in the same way, with the gap between retirement ages expected to shrink from 2.5 to 2 years.

At the same percentiles France's productivity profile is flatter than the U.S. one (table 7): it runs from 1.12 to 2.56, a ratio of the high to the low group of 2.29 against 3.08 in the U.S. (table 3). The propensity to vote is nearly flat, between 0.81 and 0.86. The common taste for leisure is 7.27 against 4.03 in the U.S., which is how the calibration accounts for France's shorter workweek.

Table 7: Household heterogeneity – France

	Income group			
Parameter	**Low**	**Medium**	**High**	**Target**
γ_i	0.58	0.18	0.24	Income percentiles.
X_i	7.27	7.27	7.27	Average hours worked.
η_i	1.12	1.58	2.56	Income distribution.
μ_i	0.81	0.86	0.85	Voting propensity.

F.3 The UK

For the UK we use the Harmonised British Household Panel Survey, wave 17, which refers to 2006–07 and is distributed by the UK Data Service.[38] Income groups are cut as for the U.S., at half-mean and mean income. The propensity to vote is from the Comparative Study of Electoral Systems.

The UK's cuts put fewer households in the low group than the U.S. cuts do, 29% against 58%, and most of them in the medium group (table 8). The ratio of high to low productivity is close to the U.S. one, 3.06 against 3.08, and the propensity to vote lies between the other two countries, rising from 0.61 to 0.74. Table 10 also reports the UK with its households regrouped at the U.S. income percentiles, by a linear fit of the UK's group means rather than a recount of the survey; the online appendix documents that grouping.

[38] Later waves were collected at least in part during the Great Recession.

Table 8: Household heterogeneity – UK

	Income group			
Parameter	**Low**	**Medium**	**High**	**Target**
γ_i	0.29	0.58	0.13	Income percentiles.
X_i	8.26	8.26	8.26	Average hours worked.
η_i	0.96	1.63	2.92	Income distribution.
μ_i	0.61	0.70	0.74	Voting propensity.

F.4 France at the UK's income groups

Since France's system is fully Bismarckian, its income groups can also be cut at the UK's own income percentiles without disturbing the identification of θ, which puts France's and the UK's household types in one-to-one correspondence exactly as the U.S. percentiles do above. Table 9 reports France's household heterogeneity on those groups; the online appendix imposes them on the UK.[39] At the UK's cuts France is the more unequal of the two: the productivity of its top group relative to its bottom group is about 3.3 against the UK's 3.06, whereas at the U.S. cuts France is well below the U.S. (table 3). The regrouping moves none of France's aggregates: its political weight of the old and its tax rate are those of table 3.

Table 9: Household heterogeneity – France, UK income groups

	Income group			
Parameter	**Low**	**Medium**	**High**	**Target**
γ_i	0.29	0.58	0.13	Income percentiles.
X_i	7.20	7.20	7.20	Average hours worked.
η_i	0.94	1.51	3.09	Income distribution.
μ_i	0.81	0.83	0.85	Voting propensity.

G Endogenous design: additional results

This appendix completes section 7: the three specifications of the cost of appendix C.2 across economies and across counterfactuals at $\rho = 1$, and the quadratic cost across intertemporal elasticities.

[39] Income, hours and group sizes are the same Luxembourg Income Study sample as table 7, regrouped. Voting propensities are only observed at the U.S. cuts; at the UK's cuts each group carries the population-weighted average of the observed values it overlaps with.

G.1 Calibration across variants and countries

Table 10 sets the three variants against the UK, France and the UK regrouped at the U.S. income percentiles, and adds the linear-quadratic cost with its shape pinned by France's corner instead of the UK's design. The own parameters say what each design asks of the cost. The UK's is within 16% of the U.S. value under the quadratic cost, 7.264 against 8.643, and 41% above it under the linear one, 2.414 against 1.710. France's is a threshold, the smallest value at which its electorate chooses the corner: no finite value under the quadratic cost, since the first unit of redistribution is free at the margin; 1.312 under the linear cost, below the U.S. value; and $\lambda \geq 9.793$ under the linear-quadratic cost at the U.S. shape, nearly twice the U.S. value. Pinning the shape by France's corner instead takes $\kappa = 0.629$, nearly five times the UK's, and a linear parameter $\lambda_1 = 1.301$, about twice the UK's; the UK then chooses 0.397 and would re-elect its design only at parameters about a fifth above the U.S. ones. The two pinnings are two points on one curve. Along the pairs (λ, κ) that keep the U.S. on its design the marginal cost of redistribution at the U.S. design is held to first order, and the marginal cost at a design θ relative to it, $[\kappa M + \tilde{V}(1-\theta)]/[\kappa M + \tilde{V}(1-\theta^*)]$ at the U.S. dispersions, falls with κ for $\theta < \theta^*$ and rises for $\theta > \theta^*$; the UK, with the U.S. ratio $M/\tilde{V}$, therefore chooses a design that falls with the shape parameter, and France, with a larger ratio, one that rises, both monotonically, so no pair on the curve puts both on their designs. Of the three economies, the weight of the linear term matters for France alone. The U.S. and the UK have nearly the same $M/\tilde{V}$, 1.32 and 1.31, so with $\tilde{V}$ in place of M on the linear term they bear nearly the same cost. France, at 1.93, then chooses 0.749, as under the quadratic cost, rather than 0.806.

Regrouping the UK at the U.S. percentiles re-reads its observed design as 0.543 and compresses its relative incomes: the dispersion $\tilde{V}$ falls from 0.223 to 0.065, the amount carried M only from 0.292 to 0.239.[40] Under the quadratic cost the choice moves little, to 0.644 against 0.625 at the UK's own cuts, since the redistributive stakes and the cost shrink together. Under a cost with a linear term it moves a long way, to 0.846 against 0.560 under the linear-quadratic cost and to the Bismarckian corner under the linear cost and under the shape pinned by France: its $M/\tilde{V}$ is 3.7 against 1.31 at its own cuts, so the linear term weighs far more on it relative to its redistributive stakes, the mechanism by which France's income distribution moves the U.S. design (section 7.3).

[40] The regrouping is a linear fit of the UK's group means on the cumulative population shares rather than a recount of the survey (appendix F), and the fit compresses the distribution. Since the UK's own low and medium groups, 29% and 58% of households, have mean incomes of half the mean and the mean, the bottom 58% average at most 0.75 of mean income and the top 24% at least 1.6, so an exact regrouping would have $\tilde{V}$ of at least 0.10 and a lower $M/\tilde{V}$, the ratio through which a linear term moves the regrouped UK's design.

Table 10: The cost specifications across economies

Economy	θ^*	**Tax rate**	$\tilde{V}$	M	θ **chosen**	**Own** λ_1	**Own** λ_2
Linear-quadratic cost, $\kappa = 0.129$ *pinned by the UK*							
U.S.	0.738	14.4%	0.436	0.575	0.738	0.674	5.237
UK	0.560	11.9%	0.223	0.292	0.560	0.674	5.237
France	1.000	21.3%	0.218	0.422	0.806	≥ 1.260	≥ 9.793
UK at U.S. income groups	0.543	11.9%	0.065	0.239	0.846	0.411	3.196
Linear cost							
U.S.	0.738	14.4%	0.436	0.575	0.738	1.710	0
UK	0.560	11.9%	0.223	0.292	0.000	2.414	0
France	1.000	21.3%	0.218	0.422	1.000	≥ 1.312	0
UK at U.S. income groups	0.543	11.9%	0.065	0.239	1.000	0.807	0
Quadratic cost							
U.S.	0.738	14.4%	0.436	0.575	0.738	0	8.643
UK	0.560	11.9%	0.223	0.292	0.625	0	7.264
France	1.000	21.3%	0.218	0.422	0.749	–	–
UK at U.S. income groups	0.543	11.9%	0.065	0.239	0.644	0	6.519
Linear-quadratic cost, $\kappa = 0.629$ *pinned by France*							
U.S.	0.738	14.4%	0.436	0.575	0.738	1.301	2.068
UK	0.560	11.9%	0.223	0.292	0.397	1.580	2.512
France	1.000	21.3%	0.218	0.422	1.000	≥ 1.302	≥ 2.070
UK at U.S. income groups	0.543	11.9%	0.065	0.239	1.000	0.674	1.072

Note: Log preferences, $\rho = 1$. One block per specification of the cost of table 4, each with its own U.S. calibration; the last block pins the linear-quadratic cost's κ by France's corner instead of the UK's design. θ^* and the tax rate are each economy's observed design and 2020 tax rate, those of table 3 except the design of the UK at U.S. income groups, which is this table's own, read off the UK workbook regrouped at the U.S. income percentiles; $\tilde{V} = \sum_i \gamma_i (y_i - 1)^2 / y_i$ and $M = \sum_i \gamma_i |y_i - 1|$ are the dispersions of relative incomes that weight the quadratic and the linear term of the cost. "θ chosen" is the design in 2020 on the economy's own freely simulated path under the block's U.S. parameters, which the U.S. re-elects by construction; "own" are the parameters at which that path re-elects the observed design, with ω recalibrated at each trial value and, under the linear-quadratic cost, $\kappa = \lambda_1 / \lambda_2$ held at the block's value. France's observed design is the corner $\theta = 1$, so its own values are the smallest at which its electorate chooses the corner ($\geq$); under the quadratic cost no finite value does (–). Each economy is its own calibration at $\rho = 1$ (β imposed from the U.S.). With $\tilde{V}$ in place of M on the linear term, at $\kappa = 0.170$ and with the U.S. design re-elected ($\lambda_1 = 0.889$, $\lambda_2 = 5.237$), France chooses 0.749 under the U.S. parameters.

G.2 Counterfactuals across variants

Table 11 reads every scenario under the three variants, with the savings rate and the workweek. Mild ageing moves the chosen design a little more than half as far as acute ageing under the linear-quadratic and the quadratic cost, to 0.811 and 0.784, and to the Bismarckian corner under the linear cost. Under the linear-quadratic cost France's income

distribution moves the design further than mild ageing does and nearly as far as acute ageing, 0.850 against 0.874, whereas under the quadratic cost it moves it less than mild ageing, 0.772 against 0.784. Choosing the design adds to the rise in the tax rate under both ageing scenarios and every variant, by at most 0.4 p.p. under the linear-quadratic cost and 1.3 p.p. under the linear one.

The savings rate and the workweek of 2020 respond to the design chosen in 2020 for the retirement of the young: a more earnings-related benefit lowers the marginal wedge of the flat component on their hours and, with less of the revenue lost, raises what they expect from the pension, so they work more and save less. On the baseline that design is above the observed one under every variant (figure 7.1), so even there the savings rate is lower with the design chosen than with it pinned and the workweek longer, by 0.1 p.p. of GDP and 0.1 hours under the linear-quadratic cost and by 0.5 p.p. of GDP and 0.4 hours under the linear cost. Ageing widens the gap in the savings rate under every variant, to 0.3 p.p. under the linear-quadratic cost and 0.7 p.p. under the linear one at acute ageing, and narrows the gap in the workweek, since the chosen design then comes with a higher tax rate. France's voting patterns, under which the electorate chooses flatter designs, reverse both gaps.

Under the linear cost France's income distribution and voting patterns, which alone send the design to a corner or next to it, together leave it at 0.743, beside the U.S. design; under the other two variants the voting patterns prevail (section 7.3). Adding France's taste for leisure shortens the workweek by about five hours under every variant and leaves the design and every rate as they are, as it does with the design given (section 6). With all of France's household characteristics the U.S. keeps a tax rate of 14.4%, against 21.3% on France's own path, and chooses 0.641 under the linear-quadratic cost, against France's 0.806 under the same parameters (table 10): with a political weight of the old no higher than the U.S. one, what sets France's system apart is its demography rather than its households. On the UK as host, at its own parameters, France's voting patterns move the design toward flat benefits under every variant and prevail over its income distribution, which leaves the UK's design within 0.01 of its observed value except under the linear cost (online appendix).

Table 11: Endogenous pension design across counterfactuals and cost specifications: all readings

		Tax rate		Savings rate		Workweek	
Scenario	**θ chosen**	**Pinned**	**Chosen**	**Pinned**	**Chosen**	**Pinned**	**Chosen**
Linear-quadratic cost							
Baseline	0.738	14.43%	14.43%	15.54%	15.41%	39.39	39.51
Mild ageing	0.811	18.08%	18.22%	−0.77 p.p.	−0.97 p.p.	39.18	39.29
Acute ageing	0.874	22.29%	22.73%	−1.76 p.p.	−2.05 p.p.	38.90	38.96
Income distribution	0.850	13.81%	13.91%	+0.28 p.p.	+0.03 p.p.	40.23	40.45
Voting patterns	0.622	15.25%	15.25%	−0.15 p.p.	−0.12 p.p.	39.21	39.20
Income distribution and voting patterns	0.641	14.41%	14.41%	+0.06 p.p.	+0.04 p.p.	40.10	40.13
All French characteristics	0.641	14.41%	14.41%	+0.06 p.p.	+0.04 p.p.	34.95	34.98
France	1.000	21.29%	–	−1.97 p.p.	–	35.24	–
Linear cost							
Baseline	0.738	14.43%	14.43%	15.61%	15.16%	39.39	39.76
Mild ageing	1.000	17.93%	18.57%	−0.41 p.p.	−1.01 p.p.	39.22	39.41
Acute ageing	1.000	21.97%	23.24%	−1.36 p.p.	−2.09 p.p.	38.99	39.02
Income distribution	1.000	13.85%	14.51%	+0.58 p.p.	−0.01 p.p.	40.22	40.42
Voting patterns	0.047	15.22%	15.01%	+0.18 p.p.	+0.44 p.p.	39.22	39.15
Income distribution and voting patterns	0.743	14.44%	14.44%	+0.37 p.p.	0.00 p.p.	40.09	40.45
All French characteristics	0.743	14.44%	14.44%	+0.37 p.p.	0.00 p.p.	34.93	35.24
France	1.000	21.29%	–	−1.72 p.p.	–	35.24	–
Quadratic cost							
Baseline	0.738	14.43%	14.43%	15.50%	15.41%	39.39	39.47
Mild ageing	0.784	18.18%	18.27%	−0.84 p.p.	−0.96 p.p.	39.15	39.22
Acute ageing	0.825	22.51%	22.80%	−1.85 p.p.	−2.04 p.p.	38.83	38.87
Income distribution	0.772	13.78%	13.80%	+0.25 p.p.	+0.16 p.p.	40.24	40.34
Voting patterns	0.653	15.27%	15.26%	−0.20 p.p.	−0.18 p.p.	39.21	39.20
Income distribution and voting patterns	0.627	14.40%	14.44%	+0.03 p.p.	+0.05 p.p.	40.10	40.06
All French characteristics	0.627	14.40%	14.44%	+0.03 p.p.	+0.05 p.p.	34.96	34.93
France	1.000	21.29%	–	−1.97 p.p.	–	35.24	–

Note: Log preferences, $\rho = 1$. One block per specification of the cost, each at its U.S. calibration (table 4). Every row is a separate equilibrium path read in 2020, with the changed characteristic holding over the whole horizon and the political choice binding from the first period. "Pinned" holds the design at θ^* and "chosen" lets the electorate choose it one period in advance, both under the cost. The ageing scenarios are those of section 6; the two composite rows impose France's income distribution and voting patterns together and, in the second, also its level of X. The France row is France's own calibrated path at its observed design $\theta = 1$, where none of the three costs bites, and has no chosen reading. The savings rate is savings relative to GDP: its level on the baseline row and, on every other row and in both columns, the change against the baseline with the design chosen, in percentage points. The workweek is in hours.

G.3 The role of the intertemporal elasticity

Tables 12 and 13 hold the quadratic cost at $\rho = 0.5$, 1 and 2. As its parameter falls with ρ, so does the share of revenue lost at the observed design, from 3.9% at $\rho = 0.5$ to 0.4% at $\rho = 2$, and at a flat benefit, from 43.7% to 5.3%. A higher elasticity strengthens the forward-looking channels that hold the choice away from the Beveridgean corner, so less cost is needed to keep the U.S. electorate at its observed design. Without any cost the same channels show in the timing of the choice: chosen with the tax, the design is the Beveridgean corner at every elasticity checked, while chosen once and for all, which enters every channel at once, it is the Beveridgean corner up to $\rho = 1.3$ and the Bismarckian corner from $\rho = 1.4$ on (online appendix).

Table 12: The calibrated cost of redistribution

CRRA (ρ)	λ	θ^*	$f(\theta^*)$	$f(0)$	$\tilde{V}$
0.5	18.267	0.738	0.961	0.563	0.436
1.0	8.643	0.738	0.982	0.762	0.436
2.0	1.724	0.738	0.996	0.947	0.436

Note: The quadratic cost, $f(\theta, \tau) = \exp\{-\frac{1}{2}\lambda\tau\tilde{V}(1-\theta)^2\}$ with $\tilde{V} = \sum_i \gamma_i(y_i - 1)^2/y_i$ the dispersion of relative labor incomes; $f(\theta^*)$ and $f(0)$ are read at the 2020 tax rate, so $1 - f$ is the share of revenue lost at the observed design and at a flat benefit; λ is calibrated so that θ_{2020}, the design chosen in 1990, is the observed one, with (β, ω) recalibrated at each trial value. The cost is proportional, so it cancels from the replacement-rate ratio and θ^* is the data's own. Without the cost the choice would be in the corner $\theta = 0$ for every ρ.

Acute ageing moves the chosen design about equally at every ρ (section 7.3), and choosing the design adds 0.1 to 0.3 p.p. to the rise in the tax rate under it. France's income distribution moves the design toward flat benefits at $\rho = 0.5$, to 0.707, and toward earnings-related ones at $\rho = 1$ and 2; its voting patterns move it toward flat benefits at every ρ, furthest at $\rho = 2$, and the composite of the two follows them, to 0.667, 0.627 and 0.262. At $\rho = 2$, where the cost is thinnest, choosing the design also raises the tax rate, by 0.7 p.p. under France's voting patterns and 0.4 p.p. under the composite, while at $\rho \leq 1$ it moves the tax rate by at most 0.1 p.p. in these scenarios. Across ρ under the quadratic cost, as across the variants at $\rho = 1$, ageing makes the chosen design more Bismarckian everywhere, while the response to inequality changes direction with ρ and size with the variant.

The UK's own quadratic cost falls less steeply with ρ than the U.S. one: 16 to 17% below it at $\rho \leq 1$, it is 61% above it at $\rho = 2$ (online appendix). The Frisch elasticity, through which the design acts on hours, matters little: recalibrated at $\xi = 0.2$ and 0.4, the cost parameter moves by about one percent and the chosen designs on the baseline path and under acute ageing by at most 0.001, while the tax rate under acute ageing

Table 13: Endogenous pension design across counterfactuals and intertemporal elasticities in the U.S.

Scenario	$\rho = 0.5$			$\rho = 1.0$			$\rho = 2.0$		
	θ **chosen**	**Tax, pinned**	**Tax, chosen**	θ **chosen**	**Tax, pinned**	**Tax, chosen**	θ **chosen**	**Tax, pinned**	**Tax, chosen**
Baseline	0.738	14.43%	14.41%	0.738	14.43%	14.43%	0.738	14.43%	14.44%
Acute ageing	0.816	23.01%	23.32%	0.825	22.51%	22.80%	0.826	21.79%	21.92%
Income distribution	0.707	13.71%	13.67%	0.772	13.78%	13.80%	0.918	13.85%	14.09%
Voting patterns	0.687	14.73%	14.62%	0.653	15.27%	15.26%	0.335	16.08%	16.73%
French characteristics	0.667	13.98%	13.96%	0.627	14.40%	14.44%	0.262	15.04%	15.45%

Note: The deadweight cost $f(\theta_t, \tau_t)$ of equation (15) is the quadratic cost, $f = \exp\{-\frac{1}{2}\lambda\tau_t\tilde{V}(1-\theta_t)^2\}$: quadratic in the implicit tax the flat component levies and scaled by the size of the system, with λ calibrated per ρ (table 12). Every row is a separate equilibrium path read in 2020, with the changed characteristic holding over the whole horizon and the political choice binding from the first period. "Pinned" holds the design at the U.S. value θ^* and "chosen" lets the electorate choose it one period in advance, both under the cost. The composite imposes France's income distribution and voting patterns together. France's level of X, which all French characteristics add, leaves every cell of the row unchanged except the last digit of the tax at $\rho = 0.5$, so up to that digit it is also the design and the tax under all French characteristics. The savings and workweek readings are in the online appendix.

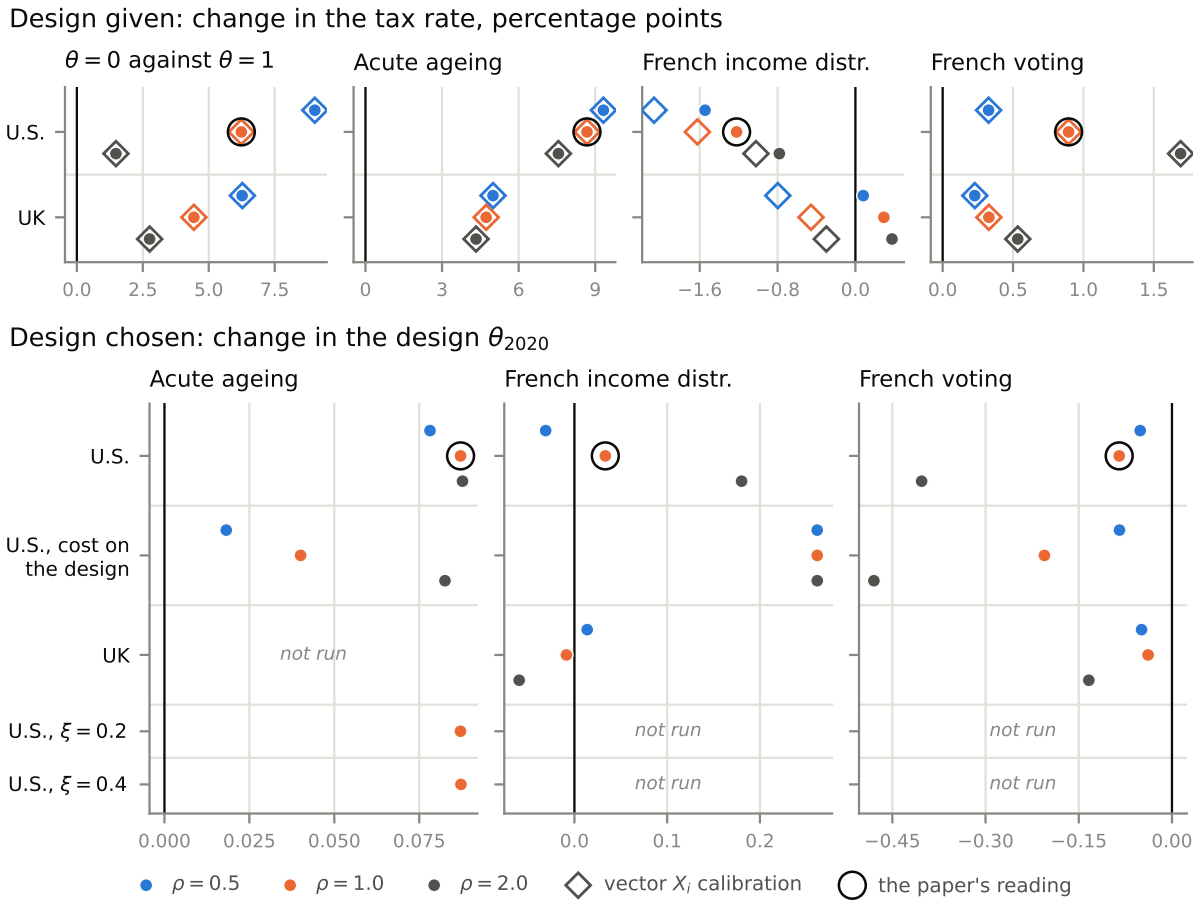
moves by about 0.4 p.p., lower at the higher elasticity (online appendix). The linear and the linear-quadratic costs are solved under log preferences only, so the comparison across ρ is the quadratic cost's.

H Robustness

Figure H.1 sets the paper's main results against the specifications behind them, one panel per result. With the pension design given, the panels are the change in the equilibrium tax rate from $\theta = 1$ to $\theta = 0$ and under acute ageing, France's income distribution and France's voting patterns, in percentage points; with the design chosen, they are the change in the chosen design under the same three counterfactuals. Each marker is one specification, read in 2020 against the host economy's own baseline at the same ρ: its colour is the intertemporal elasticity, its shape the calibration of the taste for leisure, common X or vector X_i (online appendix), and its row group the host economy. For the chosen design, which is solved under a common X only, further rows hold the alternative cost of the online appendix, the UK as host under France's characteristics and, under acute ageing at $\rho = 1$, the Frisch elasticities $\xi = 0.2$ and 0.4. The paper's own reading, the U.S. under a common X at $\rho = 1$ with the cost of section 7.1, is highlighted, and a line marks zero: the spread of a panel's markers around the highlighted one is how far the result depends on the specification, and a marker on the other side of the line is a

change of sign. The online appendix at championape.github.io/SSDclaude is organized around this figure and links each marker to the table and the row that print its value.

Figure H.1: The main results across specifications



Note: Every value is read in 2020. Table 5 prints the designs chosen for the U.S. under a common X with the cost of section 7.1 at $\rho = 1$; every other value is printed in the online appendix, sections OA.3.4 and OA.3.5 with the design given and OA.4.3 and OA.4.4 with it chosen.